SPAI

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Preface

Statistics, probability, stochastic process, machine learning structure and artificial intelligence. See order growing out of chaos, and beauty emerging from randomness.

Part I

Statistics and Probability

Chapter 1

Probability

I am a physicist, with knowledge of quantum mechanics, thus I do not quite believe in 100% certainty. Nevertheless, when dealing with daily life, classical mechanics usually suffices. Under this frame, whether an event happens or not is fully determined by the initial conditions and the laws of physics. Say that one is throwing a hexahedral dice, and in the common sense, one may say that the *probability* of each outcome equals, without exploring the detailed process. However, in eye of a Laplace's demon, the dice has gained its certainty with one hundred percent about which surface will be up, as long as the posture, texture and the initial velocity of the dice, the wind-field distribution and the texture of the contact ground are fully known, one can in principle write down the Lagrangian or Hamiltonian, list Newton's equations of motion and solve the kinematics for the dice without any error — possibly only with error for the initial condition measurements.

If we are really equipped such information and the ability of computation, each time the dice is thrown, it returns a fully determined result without ambiguity, and in this sense no term like probability will be invented. However, something, possibly disquieting or even frustrating, but also in this sense interesting, is that we do not really have such high level absorption and evaluation ability of the information, if not now then in the past, if not the simple dice problem then complex systems — there always exist situations where we are just unable to know the full historical world-line structure of an event, where what we know is just pitifully little pieces of truth, where we have to decide what to do based on unsymmetrical information. This is really the battlefield where reasoning struggles out of the wild with non-rationality.

With the unknowness of full information, the original clear world-line path splits into several distinct ones, or extends into a blurry zone, inside which the event is not determined based on known information. The event thus gains the ability to come out with distinct outcomes, and such ability for each result is called the *probability*. One may already gain some sense that the probability can change with the injection of new efficient information with clear and rational judgement, and that attaches the core of probability and statistics.

Given an arbitrary polyhedral dice, if we know it is hexahedral, then the outcomes are limited from 1 to 6, if it is further shown fair, then we shall expect and assign equal possibilities of each outcome, i.e., 1/6. This simple example illustrates how two import concepts emerges when defining probability: the *sample space* (Ω) and the *measure* (P). Basically, the former tells us the set of all possible outcomes of an *event* (E), while the latter assigns a measure (or probability) to each outcome in the sample space. Injecting new information changes the sample space and the measure, reflecting our updated

knowledge about the event.

In this chapter, nevertheless, we shall only focus on a static picture without information flowing, i.e., we are assigned with fixed sample space and measure for an event, or more precisely, we assume there exists a *random variable* X, whose possible values just fulfills the sample space Ω, in way of measurement described by P, and each time we measure it, we get a realization of the random variable as the event E. We shall utilize this known information to deduce statistical properties that we are interested in.

1.1 Common properties of probability

1.1.1 Random variables

There are basically two kinds of random variables, discrete and continuous ones. For a discrete one, we may assume that the sample space Ω is a countable set, and at many times a finite one with $\Omega = \{x_1, x_2, \ldots, x_n\}$. For each possible outcome $x_i \in \Omega$, the probability is given by $p_i = P(x_i)$, which is a positive definite map and normalized

$$\sum_{i=1}^{n} p_i = 1,\ p_i > 0. \tag{1.1}$$

The *mean* value (arithmetic average) of such a random variable X is defined as

$$\mathbb{E}[X] = \langle X \rangle = \sum_{i=1}^{n} x_i p_i, \tag{1.2}$$

which measures what value do we expect when we observe the random variable X (for many times and on average). The *variance* of such a random variable X is defined as

$$\begin{aligned} \mathrm{Var}[X] &= \mathbb{E}[(X - \mathbb{E}[X])^2] = \langle (X - \langle X \rangle)^2 \rangle = \langle X^2 \rangle - \langle X \rangle^2 \\ &= \sum_{i=1}^{n} x_i^2 p_i - \left(\sum_{i=1}^{n} x_i p_i \right)^2, \end{aligned} \tag{1.3}$$

which measures the spread or dispersion of the random variable X around its mean. The *standard deviation* of such a random variable X is defined as

$$\sigma_X = \sqrt{\mathrm{Var}[X]}. \tag{1.4}$$

There are other kinds of means, such as the geometric mean and the harmonic mean

$$\bar{x}_{\mathrm{geo}} = \left(\prod_{i=1}^{n} x_i \right)^{1/n}, \tag{1.5a}$$

$$\bar{x}_{\mathrm{har}} = \left(\sum_{i=1}^{n} \frac{1}{x_i} \right)^{-1}. \tag{1.5b}$$

The simplest example can be a coin, which has a sample space $\Omega = \{H, T\}$ with $P(H) = P(T) = 1/2$. Denoting X as the random variable representing the outcome of the coin flip, we have $X(H) = 1$ and $X(T) = 0$, and the expectation value of X is

$$\mathbb{E}[X] = \langle X \rangle = 1 \cdot P(H) + 0 \cdot P(T) = 1 \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} = \frac{1}{2}. \tag{1.6}$$

The variance of X is

$$\mathrm{Var}[X] = \langle X^2 \rangle - \langle X \rangle^2 = 1^2 \cdot P(H) + 0^2 \cdot P(T) - \left(\frac{1}{2}\right)^2 = \frac{1}{4}. \tag{1.7}$$

For continuous random variable, the sample space Ω is typically an interval or a union of intervals on the real line, and the probability is described by a probability density function (PDF) $f_X(x)$ such that

$$\int_\Omega \mathrm{d}x \; f_X(x) = 1, \;\; f_X(x) \geq 0. \tag{1.8}$$

Notice that the probability of taking any specific value reads

$$\int_{x_0-\epsilon}^{x_0+\epsilon} \mathrm{d}x \; f_X(x) = P(x_0) \to 0,$$

where $\epsilon = 0^+$ is an infinitesimally small positive number, which forces the integral to approach zero in common cases, i.e., event happens with a definite value, but the probability of any definite value to be chosen goes to zero. Nevertheless, it is interesting to observe that we may take

$$f_X(x) = \sum_{i=1}^{n} p_i \delta(x - x_i), \tag{1.9}$$

where $\delta(x)$ is the Dirac delta function, and p_i is the probability of the outcome x_i, such that the discrete case can be unified under the continuous framework. The mean value

$$\mathbb{E}[X] = \langle X \rangle = \int_\Omega \mathrm{d}x \; x f_X(x). \tag{1.10}$$

The variance of such a random variable X is defined as

$$\begin{aligned} \mathrm{Var}[X] &= \mathbb{E}[(X - \mathbb{E}[X])^2] = \langle (X - \langle X \rangle)^2 \rangle = \langle X^2 \rangle - \langle X \rangle^2 \\ &= \int_\Omega \mathrm{d}x \; (x - \mathbb{E}[X])^2 f_X(x). \end{aligned} \tag{1.11}$$

For continuous variables, we have cumulative distribution:

$$F_X(a) = P(X \leq a) = \int_{-\infty}^{a} \mathrm{d}x \; f_X(x), \tag{1.12}$$

which relates to the PDF by derivative

$$f_X(x) = \frac{\mathrm{d}}{\mathrm{d}x} F_X(x). \tag{1.13}$$

Possibly the simplest example of a continuous random variable is the uniform distribution, where the PDF is constant over a given interval $f_X(x) = \frac{1}{W}$ for $x \in [a, a+W]$ and zero otherwise. The expectation value of the distribution reads

$$\begin{aligned} \mathbb{E}[X] = \langle X \rangle &= \int_a^{a+W} \mathrm{d}x \; x f_X(x) = \frac{1}{W} \int_a^{a+W} \mathrm{d}x \; x \\ &= \frac{1}{W} \left[\frac{x^2}{2} \right]_a^{a+W} = \frac{(a+W)^2 - a^2}{2W} = a + \frac{W}{2}, \end{aligned} \tag{1.14}$$

as expected, that it should fall onto the middle point of the interval. The variance of the distribution reads

$$\begin{aligned}\mathrm{Var}[X] = \langle X^2\rangle - \langle X\rangle^2 &= \int_a^{a+W} \mathrm{d}x\ x^2 f_X(x) - \left(a+\frac{W}{2}\right)^2 \\ &= \frac{1}{W}\int_a^{a+W} \mathrm{d}x\ x^2 - \left(a+\frac{W}{2}\right)^2 \\ &= \frac{1}{W}\left[\frac{x^3}{3}\right]_a^{a+W} - \left(a+\frac{W}{2}\right)^2 \\ &= \frac{(a+W)^3-a^3}{3W} - \left(a+\frac{W}{2}\right)^2 = \frac{W^2}{12},\end{aligned} \tag{1.15}$$

irrelevant with the end points of the interval.

There are several important properties of expectation values and variances

$$\mathbb{E}(aX+b) = a\mathbb{E}(X) + b, \tag{1.16a}$$
$$\mathbb{E}(X_1+X_2) = \mathbb{E}(X_1) + \mathbb{E}(X_2), \tag{1.16b}$$
$$\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X), \tag{1.16c}$$

with the first two seen by linearity and the last seen by knowing the variance as basically square of variable. A straightforward derivation also suffices

$$\begin{aligned}\mathbb{E}(aX+b) &= \sum_{i=1}^n (ax_i+b)p_i = a\sum_{i=1}^n x_i p_i + b\sum_{i=1}^n p_i = a\mathbb{E}(X)+b, \\ \mathbb{E}(X_1+X_2) &= \sum_{i=1}^n (x_1^i+x_2^i)p_i = \sum_{i=1}^n x_1^i p_i + \sum_{i=1}^n x_2^i p_i = \mathbb{E}(X_1)+\mathbb{E}(X_2), \\ \mathrm{Var}(aX+b) &= \langle (aX+b)^2\rangle - \langle aX+b\rangle^2 \\ &= \langle a^2X^2 + 2abX + b^2\rangle - (a\langle X\rangle + b)^2 \\ &= a^2\langle X^2\rangle + 2ab\langle X\rangle + b^2 - (a^2\langle X\rangle^2 + 2ab\langle X\rangle + b^2) \\ &= a^2(\langle X^2\rangle - \langle X\rangle^2) = a^2\mathrm{Var}(X).\end{aligned}$$

These two formulas show that the expectation operation is a linear map, while the variance does not change with the center of the random variable.

For continuum, there are also cases with change of measure, or equivalently $x \to y(x)$, and at this time, by using normalization condition

$$1 = \int \mathrm{d}y\ f_Y(y) = \int \mathrm{d}x\ y'(x) f_Y(y(x)) = \int \mathrm{d}x\ f_X(x), \tag{1.17}$$

one has

$$f_Y(y) = \frac{1}{y'(x)} f_X(x) \quad \text{where } y = y(x). \tag{1.18}$$

Notice that this assumed monotonous property of $y(x)$, and when it is not the case, one needs to split the domain of integration into monotonous zones, which is equivalent to calculating the cumulant probability first. People also at many times meet with condition to get the expectation value of some function $g(X)$, and that gives

$$\langle g(X)\rangle = \int_\Omega \mathrm{d}x\ g(x) f_X(x). \tag{1.19}$$

1.1.2 Joint distribution

There are cases where we count simultaneously several random variables, forming a joint probability distribution. In smallest case, there are two variables, and for discrete case,

$$f_{X,Y}(x_i, y_j) = P(X = x_i, Y = y_j), \tag{1.20}$$

with $P(X = x_i, Y = y_j)$ standing for the probability that both $X = x_i$ and $Y = y_j$ occur. For continuous variables, we firstly have the joint cumulative distribution function:

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y). \tag{1.21}$$

The joint probability density function $f_{X,Y}(x, y)$ is the second partial derivative of the joint cumulative distribution function:

$$f_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y), \tag{1.22}$$

which comes from the definition

$$P(X \leq x, Y \leq y) = \int_{-\infty}^{x} \int_{-\infty}^{y} \mathrm{d}u\mathrm{d}v \; f_{X,Y}(u, v). \tag{1.23}$$

One is able to obtain the marginal distributions by integrating out one of the variables:

$$f_X(x_i) = \sum_j f_{X,Y}(x_i, y_j), \tag{1.24a}$$

$$f_X(x) = \int_{-\infty}^{\infty} \mathrm{d}y \; f_{X,Y}(x, y), \tag{1.24b}$$

and similarly for $f_Y(y_{(i)})$. It is then straightforward to obtain mean and variance for each marginal distribution.

With the joint PDF, people may want to obtain the probability of a specific event involving both variables. Say that in reality, X and Y are constrained by the relation $xy \leq z$, with $\Omega_X = [0, u_X]$, $\Omega_Y = [0, u_Y]$, then we may want to solve

$$P(XY \leq z) = \iint_{xy \leq z} \mathrm{d}x\mathrm{d}y \; f_{X,Y}(x, y) = \int_0^{u_X} \mathrm{d}x \int_0^{z/u_X} \mathrm{d}y \; f_{X,Y}(x, y). \tag{1.25}$$

Other cases follow similarly by writing down the constraint explicitly.

If these two variables are independent, then

$$f_{X,Y}(x, y) = f_X(x) f_Y(y). \tag{1.26}$$

Meaning that the measurement of one random variable will not affect the other.

The above statements are easily generalizable to multi-variable cases.

1.1.3 Correlation and covariance

At many times, people want to know whether the change of one variable is related to the change of another variable. This relationship is quantified by the covariance and correlation coefficients. The covariance of two random variables is defined as

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]. \tag{1.27}$$

It shares several important properties as interchangeability and linearity:

$$\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X), \tag{1.28a}$$
$$\mathrm{Cov}(aX+c, bY+d) = ab\mathrm{Cov}(X,Y), \tag{1.28b}$$
$$\mathrm{Cov}(X_1+X_2, Y) = \mathrm{Cov}(X_1,Y) + \mathrm{Cov}(X_2,Y). \tag{1.28c}$$

Based on the covariance, we may define the correlation coefficient

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y} \in [-1,1], \tag{1.29}$$

which satisfies

$$\rho_{aX+c,bY+d} = \mathrm{sgn}(ab)\rho_{XY}, \tag{1.30}$$

as the standard derivation is always positive, while the covariance changes the sign.

For two independent variables X and Y, we have $f_{X,Y}(x,y) = f_X(x)f_Y(y)$, thus

$$\mathbb{E}(XY) = \sum_{x,y} xy f_{X,Y}(x,y) = \sum_x x f_X(x) \sum_y y f_Y(y) = \mathbb{E}(X)\mathbb{E}(Y).$$

This leads to $\rho_{XY} = 0$ for independent variables.

On the other hand, for linearly dependent variables $Y = aX + b$, the joint PDF does not share a usual expression but $f_{X,Y}(x,y) = f_X(x)\delta(y - ax - b)$, and we have

$$\mathbb{E}(XY) = \sum_{x,y} xy f_{X,Y}(x,y) = \sum_x x(ax+b) f_X(x) = a\left\langle X^2 \right\rangle + b\left\langle X \right\rangle,$$

which leads to

$$\mathrm{Cov}(X,Y) = a\left\langle X^2 \right\rangle + b\left\langle X \right\rangle - \left\langle X \right\rangle \left(a\left\langle X \right\rangle + b\right) = a(\left\langle X^2 \right\rangle - \left\langle X \right\rangle^2) = a\sigma_X^2,$$

and their correlation becomes

$$\rho_{XY} = \frac{a\sigma_X^2}{\sigma_X \sigma_Y} = a\frac{\sigma_X}{\sigma_Y} = a\frac{\sigma_X}{|a|\sigma_X} = \mathrm{sgn}(a). \tag{1.31}$$

When $a > 0$, we have $\rho_{XY} = 1$ and positive correlation, while when $a < 0$, we have $\rho_{XY} = -1$ and negative correlation.

Furthermore, there exists cases where one variable is composed of a sum of a series of independent and identical variables, $X = \sum_{i=1}^n Y_i$, where $\mathbb{E}(Y_i) = \mathbb{E}(Y)$ and $\mathrm{Var}(Y_i) = \mathrm{Var}(Y)$ are the same for all i. Independence further means that

$$f_{Y_i,Y_j}(y_i,y_j) = f_{Y_i}(y_i)f_{Y_j}(y_j),\ \forall i \neq j,$$

which gives $\mathrm{Cov}(Y_i,Y_j) = \mathrm{Var}(Y)\delta_{ij}$. In this case, we have the expectation value of X as

$$\mathbb{E}(X) = \mathbb{E}(\sum_{i=1}^n Y_i) = \sum_{i=1}^n \mathbb{E}(Y_i) = n\mathbb{E}(Y). \tag{1.32}$$

To calculate the variance of X, we need to firstly introduce another important property of variance,

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X,Y), \tag{1.33}$$

which can be done by

$$\begin{aligned}\mathrm{Var}(X+Y) &= \langle (X+Y-\langle X\rangle-\langle Y\rangle)^2\rangle \\ &= \langle (X-\langle X\rangle)^2\rangle + \langle (Y-\langle Y\rangle)^2\rangle + 2\langle (X-\langle X\rangle)(Y-\langle Y\rangle)\rangle \\ &= \mathrm{Var}(X)+\mathrm{Var}(Y)+2\mathrm{Cov}(X,Y).\end{aligned}$$

Then we have

$$\begin{aligned}\mathrm{Var}(X) = \langle (X-\langle X\rangle)^2\rangle &= \langle \left(\sum_i Y_i - \langle Y_i\rangle\right)^2 \rangle \\ &= \sum_{i,j}\langle (Y_i-\langle Y\rangle_i)(Y_j-\langle Y\rangle_j)\rangle \\ &= \sum_i \langle (Y_i-\langle Y\rangle_i)^2\rangle + \sum_{i\neq j}\langle (Y_i-\langle Y\rangle_i)(Y_j-\langle Y\rangle_j)\rangle \\ &= \sum_i \mathrm{Var}(Y_i) + \sum_{i\neq j}\mathrm{Cov}(Y_i,Y_j) \\ &= n\mathrm{Var}(Y)+0 = n\mathrm{Var}(Y).\end{aligned} \tag{1.34}$$

This can also be directly deduced from the above two-variable case: that when there is no correlation in between, their variance just follows an addition.

1.1.4 Conditional probability

The linearly dependent variables indeed represent one kind of the conditional property, that with $Y = aX+b$, and $f_{X,Y}(x,y) = f_X(x)\delta(y-ax-b)$, we see that when x is measured to be x_0, then the probability that $y = ax_0+b$ is 1, and otherwise 0. This is an extreme case of conditional probability, and the form indeed enlightens us to write down

$$f_{X,Y}(x,y) = f_X(x)f_{Y|X}(y|x), \tag{1.35}$$

with $f_{Y|X}(y|x)$ being the conditional probability density function of Y given $X = x$.

This sounds a bit abstract, but we can indeed start from discrete variables, that the probability of event A happens when event B happens can be written as

$$P(A|B) = \frac{P(A,B)}{P(B)}, \tag{1.36}$$

provided that $P(B) > 0$. This is self-evident by considering that we want to see probability of $A = A_i$ when $B = B_j$, out of their sample space $\{A_i, B_j\}_{i=1,j=1}^{i=n,j=m}$, and $P(B_j) = \sum_{i=1}^n P(A_i,B_j)$ contains all the cases where $B = B_j$. Based on this case, the unnormalized probability that $A = A_i$ given $B = B_j$ is $P(A_i,B_j)$, and by normalization, we get the conditional probability.

As one can see, the term 'condition' is indeed changing the sample space of one variable based on the outcome of the other. As a simple example, we may compare

- What is the probability of throwing 1 for a dice (assumed hexahedral)?
- What is the probability of throwing 1 for a dice upon knowing the thrown number is odd?

- What is the probability of throwing 1 for a dice upon knowing the thrown number is even?

As one may have, based on the common sense, guessed that the answers are 1/6, 1/3, and 0 respectively. This, however, has a solid conditional probability reasoning, that we have

$$P(1|\text{odd}) = \frac{P(1,\text{odd})}{P(\text{odd})} = \frac{P(1)}{P(\text{odd})} = \frac{1/6}{3/6} = \frac{1}{3},$$
$$P(1|\text{even}) = \frac{P(1,\text{even})}{P(\text{even})} = \frac{0}{3/6} = 0.$$

Take advantage of this opportunity, we would like to introduce more about some basic concepts in probability. Before, we take event E as a realization of random variable X, which sounds like a single outcome, but an event can indeed be any subset of the sample space Ω, just like the thrown odd result. This indeed means that it can either be empty, aka impossible event, or the entire sample space, aka certain event, thus we have basics

$$0 \leq P(E) \leq 1,\ P(\varnothing) = 0,\ P(\Omega) = 1. \tag{1.37}$$

Beside this basic property, we can also define the complement of an event, say the complement of throwing an odd number would be an even number outcome, that

$$P(\bar{E}) = 1 - P(E),\ E \cup \bar{E} = \Omega,\ E \cap \bar{E} = \varnothing, \tag{1.38}$$

with $\cup$ and $\cap$ denoting the union and intersection of sets, respectively. For two events A and B, we have probability of union

$$P(A \cup B) = P(A) + P(B) - P(A \cap B). \tag{1.39}$$

To prove this, we need to introduce the third axiom of probability by Kolmogorov: the assumption of σ-additivity: any countable sequence of disjoint sets (synonymous with mutually exclusive events) $E_1, E_2, \ldots$ satisfies

$$P(\cup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} P(E_i),\ \text{given that } E_i \cap E_j = \varnothing,\ \forall i \neq j. \tag{1.40}$$

This property can be inherited by finite series by letting $E_{i>n} = \varnothing$ for some n. Thus, we can partition, based on the set theory, that

$$A \cup B = (A - A \cap B) \cup (A \cap B) \cup (B - A \cap B), \tag{1.41}$$

and we have

$$\begin{aligned} P(A \cup B) &= P((A - A \cap B) \cup (A \cap B) \cup (B - A \cap B)) \\ &= P(A - A \cap B) + P(A \cap B) + P(B - A \cap B) \\ &= P(A) - P(A \cap B) + P(A \cap B) + P(B) - P(A \cap B) \\ &= P(A) + P(B) - P(A \cap B), \end{aligned}$$

where in the third equality, we again utilized that

$$A = (A - A \cap B) \cup (A \cap B) \Rightarrow P(A) = P(A - A \cap B) + P(A \cap B), \tag{1.42}$$

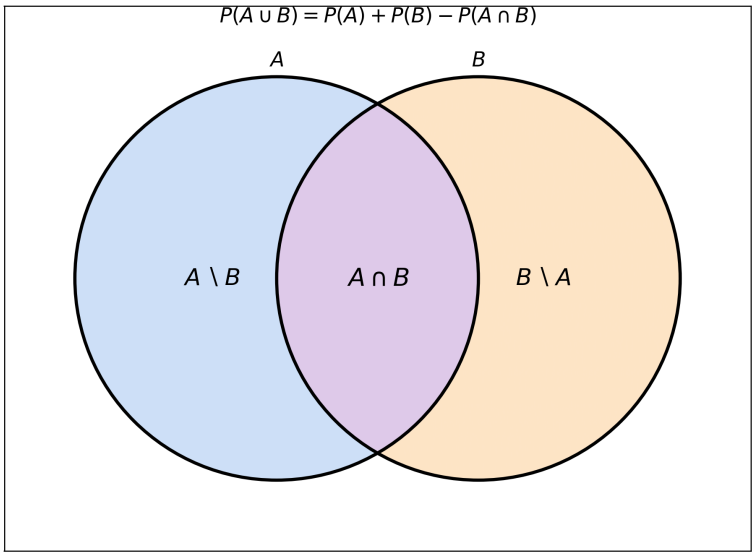


Figure 1.1: Venn diagram for the union of two events.

from which we indeed see that we can directly write

$$P(A - B) = P(A) - P(B) \text{ for } B \subseteq A. \tag{1.43}$$

Also, we have probability of intersection, by the conditional probability that

$$P(A \cap B) \equiv P(A, B) = P(A|B)P(B) = P(B|A)P(A), \tag{1.44}$$

where we have utilized the commutative law $A \cap B = B \cap A$. By noting $A \backslash B = A - A \cap B$, the Venn diagram of the partition of $A \cup B$ is shown in Fig. **??**. This formula leads to the famous *Bayes theorem*:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}, \tag{1.45}$$

and if we assume a disjoint partition of sample space

$$\cup_{i=1}^{n} A_i = \Omega, \ A_j \cap A_j = \varnothing, \ \forall i \neq j, \tag{1.46}$$

we can partition

$$B = B \cap \Omega = B \cap \cup_{i=1}^{n} A_i = \cup_{i=1}^{n} B \cap A_i, \tag{1.47}$$

we thus obtain the *partition theorem*:

$$P(B) = \sum_{i=1}^{n} P(B \cap A_i) = \sum_{i=1}^{n} P(B|A_i)P(A_i), \tag{1.48}$$

and have

$$P(A_1|B) = \frac{P(B|A_1)P(A_1)}{P(B)} = \frac{P(B|A_1)P(A_1)}{\sum_{i=1}^{n} P(B|A_i)P(A_i)}. \tag{1.49}$$

For three events union A, B and C, we similarly have

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C), \tag{1.50}$$

by partition

$$\begin{aligned}A\cup B\cup C=&(A-A\cap B-A\cap C+A\cap B\cap C)\cup(B-A\cap B-B\cap C+A\cap B\cap C)\\&\cup(C-A\cap C-B\cap C+A\cap B\cap C)\cup(A\cap B-A\cap B\cap C)\cup(A\cup C-A\cap B\cap C)\\&\cup(B\cap C-A\cap B\cap C)\cup A\cap B\cap C.\end{aligned}$$

And

$$P(A\cap B\cap C)=P(A|B\cap C)P(B\cap C)=P(A|B\cap C)P(B|C)P(C). \tag{1.51}$$

Indeed, one may easily generalize that

$$P(\cup_{i=1}^n A_i)=\sum_{i=1}^n P(A_i)-\sum_{i<j}P(A_i\cap A_j)+\sum_{i<j<k}P(A_i\cap A_j\cap A_k)-\cdots+(-1)^{n+1}P(\cap_{i=1}^n A_i), \tag{1.52}$$

together with evaluation of chain of events

$$P(\cap_{i=1}^n A_i)=\prod_{i=1}^n P(A_i|\cap_{j=1}^{i-1}A_j).$$

For continuous variables, we may formally write down the probability of $Y\in U\subseteq\Omega_Y$ when $X=x_0$ by a limit taking

$$\begin{aligned}P(Y\in U|X=x_0)&=\lim_{\varepsilon\to0}P(Y\in U|x_0-\varepsilon<X<x_0+\varepsilon)\\&=\lim_{\varepsilon\to0}\frac{\int_{x_0-\varepsilon}^{x_0+\varepsilon}\mathrm{d}x\int_U\mathrm{d}y\ f_{X,Y}(x,y)}{\int_{x_0-\varepsilon}^{x_0+\varepsilon}\mathrm{d}x\int_{\Omega_Y}\mathrm{d}y\ f_{X,Y}(x,y)}\\&=\lim_{\varepsilon\to0}\frac{\int_{x_0-\varepsilon}^{x_0+\varepsilon}\mathrm{d}x\int_U\mathrm{d}y\ f_{X,Y}(x,y)}{\int_{x_0-\varepsilon}^{x_0+\varepsilon}\mathrm{d}x\ f_X(x)}\\&=\frac{\int_U\mathrm{d}y\ f_{X,Y}(x_0,y)}{f_X(x_0)},\end{aligned} \tag{1.53}$$

and a further derivative illustrates that we can define a conditional probability density distribution

$$f_{Y|X}(y|x)=\frac{f_{X,Y}(x,y)}{f_X(x)}. \tag{1.54}$$

Notice that when X and Y are independent, $f_{Y|X}(y|x)=f_Y(y)$, which can either be called the marginal PDF or just the original PDF.

Also notice that the conditional probability itself serves as a density distribution as

$$\sum_y f_{Y|X}(y|x),\quad\forall x\in\Omega_X, \tag{1.55}$$

with a simplified notation for the sum. This allows us to define the so-called conditional expectation value

$$\mathbb{E}(Y|X=x)=\sum_y y\cdot f_{Y|X}(y|x),\quad\forall x\in\Omega_X, \tag{1.56}$$

and the conditional variance

$$\begin{aligned}\mathrm{Var}(Y|X=x) &= \sum_y (y-\mathbb{E}(Y|X=x))^2 \cdot f_{Y|X}(y|x) \\ &= \mathbb{E}(Y^2|X=x) - (\mathbb{E}(Y|X=x))^2, \quad \forall x \in \Omega_X.\end{aligned} \tag{1.57}$$

This indeed enlightens us to re-write the expectation of Y given joint PDF $f_{X,Y}$ as

$$\begin{aligned}\mathbb{E}(Y) &= \sum_y f_Y(y) = \sum_y \sum_x f_{X,Y}(x,y) = \sum_x f_X(x) \sum_y y f_{Y|X}(y|x) \\ &= \sum_x f_X(x)\mathbb{E}(Y|X=x) = \mathbb{E}_X[\mathbb{E}(Y|X)].\end{aligned} \tag{1.58}$$

More generally, one can easily show that

$$\mathbb{E}(g(Y)) = \mathbb{E}_X[\mathbb{E}(g(Y)|X)]. \tag{1.59}$$

This may be written using double bracket as

$$\langle g(Y)\rangle = \langle\langle g(Y)\rangle_{Y|X}\rangle_X . \tag{1.60}$$

The variance also shares such kind of expressions

$$\begin{aligned}\mathrm{Var}(Y) &= \langle Y^2\rangle - (\langle Y\rangle)^2 = \langle\langle Y^2\rangle_{Y|X}\rangle_X - (\langle\langle Y\rangle_{Y|X}\rangle_X)^2 \\ &= \langle\langle Y^2\rangle_{Y|X}\rangle_X - \langle(\langle Y\rangle_{Y|X})^2\rangle_X + \langle(\langle Y\rangle_{Y|X})^2\rangle_X - (\langle\langle Y\rangle_{Y|X}\rangle_X)^2 \\ &= \langle \mathrm{Var}(Y|X)\rangle_X + \mathrm{Var}_X[\langle Y\rangle_{Y|X}]_X \\ &= \mathbb{E}_X[\mathrm{Var}(Y|X)] + \mathrm{Var}_X[\langle Y\rangle_{Y|X}].\end{aligned} \tag{1.61}$$

1.1.5 Generating functions

For $\Omega = \{0, 1, 2, \ldots\}$, we may define the *probability generating function* (PGF)

$$G_X(s) = \mathbb{E}(s^X) = \sum_{k=0}^{\infty} s^k P(X=k). \tag{1.62}$$

Two self-evident properties are

$$G_X(0) = P(X=0),\ G_X(1) = 1. \tag{1.63}$$

To generate the probability distribution, we act by derivatives

$$\begin{aligned}\left.\frac{\partial G_X(s)}{\partial s}\right|_{s=0} &= \left.\sum_{k=1}^{\infty} k s^{k-1}\right|_{s=0} P(X=k) = P(X=1) \\ \left.\frac{\partial^2 G_X(s)}{\partial s^2}\right|_{s=0} &= 2P(X=2) \\ &\cdots \\ \left.\frac{\partial^n G_X(s)}{\partial s^n}\right|_{s=0} &= n!P(X=n), \quad n \geq 1.\end{aligned}$$

This leads to

$$P(X=n)=\frac{1}{n!}\frac{\partial^n G_X(s)}{\partial s^n}\bigg|_{s=0},\quad n\geq 0. \tag{1.64}$$

Thus, we can represent the expectation value by the generating function as

$$\begin{aligned}\mathbb{E}(X)&=\sum_{n=0}^{\infty}n\frac{1}{n!}\frac{\partial^n G_X(s)}{\partial s^n}\bigg|_{s=0}=\sum_{n=1}^{\infty}\frac{1}{(n-1)!}\frac{\partial^{n-1}}{\partial s^{n-1}}\frac{\partial}{\partial s}G_X(s)\bigg|_{s=0}\\&=e^{\partial/\partial s}\frac{\partial}{\partial s}G_X(s)\bigg|_{s=0}=G'_X(s)\big|_{s=1}.\end{aligned} \tag{1.65}$$

Or more directly,

$$G'_X(s)\big|_{s=1}=\sum_{k=0}^{\infty}ks^{k-1}P(X=k)\big|_{s=1}=\sum_{k=0}^{\infty}kP(X=k)=\mathbb{E}(X).$$

Similarly, we have

$$\mathbb{E}(X(X-1)\cdots(X-n+1))=\frac{\partial^n G_X(s)}{\partial s^n}\bigg|_{s=1}. \tag{1.66}$$

We thus have

$$\begin{aligned}\mathrm{Var}(X)&=\mathbb{E}(X^2)-(\mathbb{E}(X))^2\\&=\mathbb{E}(X^2-X)+\mathbb{E}(X)-(\mathbb{E}(X))^2\\&=\left[\frac{\partial^2 G_X(s)}{\partial s^2}+\frac{\partial G_X(s)}{\partial s}\left(1-\frac{\partial G_X(s)}{\partial s}\right)\right]\bigg|_{s=1}.\end{aligned} \tag{1.67}$$

There exists a theorem by Abel that the convergence radius R of $G_X(s)$ satisfies $R\geq 1$.

As $\mathbb{E}(s^{X_1+X_2})=\mathbb{E}(s^{X_1}s^{X_2})$, we have that for $Y=\sum_{i=1}^{n}X_i$,

$$G_Y(s)=G_{X_1}(s)G_{X_2}(s)\cdots G_{X_n}(s), \tag{1.68}$$

where X_i are independent random variables.

We may let $s=e^t$ for $t\in\mathbb{R}$, thus obtaining the *moment generating function* (MGF)

$$M_X(t)=\mathbb{E}(e^{tX})=G_X(e^t). \tag{1.69}$$

This function generates the moment of all orders as

$$m_n=\mathbb{E}(X^n)=M_X^{(n)}(0), \tag{1.70}$$

since

$$M_X^{(n)}(0)=\frac{\mathrm{d}^n}{\mathrm{d}t^n}\mathbb{E}(e^{tX})\bigg|_{t=0}=\sum_x p(x)\frac{\mathrm{d}^n}{\mathrm{d}t^n}e^{tx}\bigg|_{t=0}=\sum_x p(x)x^n.$$

This indeed means that

$$M_X(t)=\sum_{k=0}^{\infty}m_k\frac{t^k}{k!}, \tag{1.71}$$

as a Taylor series, which can be used in practical applications. Note that for independent variables, we also have

$$M_{X_1+X_2}(t)=M_{X_1}(t)M_{X_2}(t). \tag{1.72}$$

If we perform analytic continuation of MGF to

$$\varphi(t) = M_X(it) = \mathbb{E}(e^{itX}), \tag{1.73}$$

we obtain the *characteristic function*. Notice that since

$$\varphi(t) = \int \mathrm{d}x\ f_X(x) e^{itx},$$

the characteristic function is indeed the Fourier transformation of the original PDF (for undefined x, just let $f_X(x) = 0$), and we can restore the PDF by inverse Fourier transformation as

$$f_X(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \varphi(t) e^{-itx} \mathrm{d}t. \tag{1.74}$$

We can also obtain PDF from MGF using inverse Laplace transformation as

$$f_X(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} M_X(t) e^{tx} \mathrm{d}t, \tag{1.75}$$

where c is a real number such that the integral converges.

Sometimes, we want to calculate the cumulants, instead of the moments of a distribution. And that comes to the *cumulant generating function* (CGF):

$$K_X(t) = \ln M_X(t) = \ln \mathbb{E}(e^{tX}). \tag{1.76}$$

The n-th cumulant is given by

$$\kappa_n = K_X^{(n)}(0) = \frac{\mathrm{d}^n}{\mathrm{d}t^n} \ln \mathbb{E}(e^{tX}) \bigg|_{t=0}, \tag{1.77}$$

or it shows the Taylor expansion:

$$\ln M_X(t) = \sum_{n=1}^{\infty} \frac{\kappa_n}{n!} t^n,\ \ln \varphi(t) = \sum_{n=1}^{\infty} \frac{\kappa_n}{n!} (it)^n. \tag{1.78}$$

From the comparison of MGF and CGF, we indeed have

$$\exp\left(\sum_{n=1}^{\infty} \frac{\kappa_n}{n!} t^n \right) = \sum_{k=0}^{\infty} m_k \frac{t^k}{k!}.$$

This gives the relation between moments and cumulants by

$$m_n = \sum_{k=1}^{n} B_{n,k}(\kappa_1, \kappa_2, \ldots, \kappa_{n-k+1}), \tag{1.79a}$$

$$\kappa_n = m_n - \sum_{k=1}^{n-1} \binom{n-1}{k-1} \kappa_k m_{n-k},\ \kappa_1 = m_1. \tag{1.79b}$$

For instance, we have

$$\kappa_1 = m_1, \tag{1.80a}$$

$$\kappa_2 = m_2 - m_1^2, \tag{1.80b}$$

$$\kappa_3 = m_3 - 3m_2 m_1 + 2m_1^3, \tag{1.80c}$$

$$\kappa_4 = m_4 - 4m_3 m_1 - 3m_2^2 + 12 m_2 m_1^2 - 6m_1^4. \tag{1.80d}$$

In physics, the cumulants can represent the correlation functions.

1.2 Common distributions

1.2.1 Discrete variables

Let us start from the coin, which is a simple example of a **Bernoulli distribution**. The sample space of Bernoulli distribution contains two possibilities

$$f_X(a) = \begin{cases} p & \text{if } a = 1, \\ 1-p & \text{if } a = 0, \end{cases} \tag{1.81}$$

then the expectation and variance of X are

$$\mathbb{E}(X) = p \cdot 1 + (1-p) \cdot 0 = p, \tag{1.82a}$$
$$\mathrm{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = p - p^2 = p(1-p). \tag{1.82b}$$

The generating function for the Bernoulli distribution is

$$G_X(s) = p \cdot s^1 + (1-p) \cdot s^0 = ps + (1-p), \tag{1.83}$$

and we have

$$G_X(0) = 1 - p, \ \frac{\partial}{\partial s} G_X(0) = p.$$

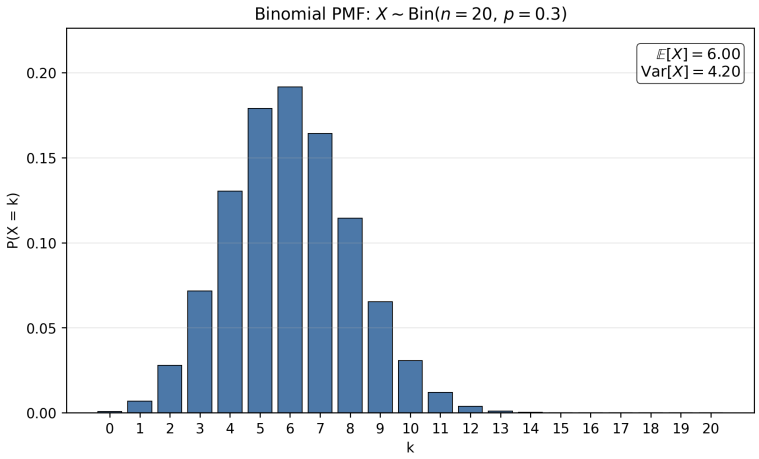


Figure 1.2: Probability mass function of a binomial distribution.

A **binomial distribution** further refers to n-sequential Bernoulli events, that $X = b(n,p) = \sum_{i=1}^{n} X_i$, with sample space $\{0, 1, 2, \ldots, n\}$. By simple permutation and combination, we have

$$f_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n. \tag{1.84}$$

The distribution is normalized by the binomial theorem. By the independence of events, we can directly write

$$\mathbb{E}(X) = \sum_{i=1}^{n} \mathbb{E}(X_i) = np, \tag{1.85a}$$
$$\mathrm{Var}(X) = \sum_{i=1}^{n} \mathrm{Var}(X_i) = np(1-p). \tag{1.85b}$$

On the other hand, we can also calculate by brute force as

$$\begin{aligned}
\mathbb{E}(X) &= \sum_{k=0}^{n} k f_X(k) = \sum_{k=1}^{n} k \binom{n}{k} p^k (1-p)^{n-k} \\
&= \sum_{k=1}^{n} \frac{n!}{(k-1)!(n-k)!} p^k (1-p)^{n-k} \\
&= np \cdot \sum_{k=1}^{n} \frac{(n-1)!}{(k-1)!(n-k)!} p^{k-1} (1-p)^{n-k} = np\,(p+(1-p))^{n-1} \\
&= np, \\
\mathrm{Var}(X) &= \mathbb{E}(X^2) - (\mathbb{E}(X))^2 \\
&= \sum_{k=1}^{n} k^2 \binom{n}{k} p^k (1-p)^{n-k} - (np)^2 \\
&= \sum_{k=1}^{n} k \frac{n!}{(k-1)!(n-k)!} p^k (1-p)^{n-k} - (np)^2 \\
&= \sum_{k=1}^{n} [1+(k-1)] \frac{n!}{(k-1)!(n-k)!} p^k (1-p)^{n-k} - (np)^2 \\
&= np \sum_{k=1}^{n} \frac{(n-1)!}{(k-1)!(n-k)!} p^k (1-p)^{n-k} \\
&+ n(n-1)p^2 \sum_{k=2}^{n} \frac{(n-2)!}{(k-2)!(n-k)!} p^{k-2} (1-p)^{n-k} - (np)^2 \\
&= np + n(n-1)p^2 - (np)^2 \\
&= np(1-p).
\end{aligned}$$

The generating function of $b(n,p)$ is

$$G_X(s) = (ps + 1 - p)^n, \tag{1.86}$$

as a product of sequential Bernoulli trials, which leads to probability distribution as

$$\begin{aligned}
f_X(k) &= \frac{1}{k!} \frac{\mathrm{d}^k}{\mathrm{d}s^k} G_X(s) \Big|_{s=0} \\
&= \frac{1}{k!} n(n-1)\cdots(n-k+1) p^k (1-p)^{n-k} = \binom{n}{k} p^k (1-p)^{n-k}.
\end{aligned} \tag{1.87}$$

Sometimes, we meet case where the happening probability p of a single event goes to zero, and at the same time the number of happening events n goes to infinity, while the product of the two $\lambda = np$ remains constant. This is the scenario for the **Poisson**

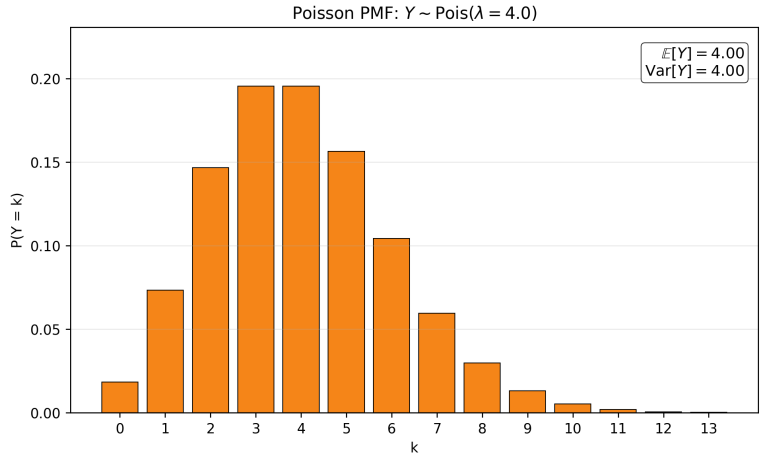


Figure 1.3: Probability mass function of a Poisson distribution.

distribution: $X = \lim_{n\to\infty} b(n, \lambda/n)$, that we have

$$\begin{aligned} f_X(k) &= \lim_{n\to\infty} \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1-\frac{\lambda}{n}\right)^{n-k} \\ &= \lim_{n\to\infty} \frac{n!}{n^k k!(n-k)!} \lambda^k \left(1-\frac{\lambda}{n}\right)^{n-k} \\ &= \lim_{n\to\infty} \frac{1}{k!} \frac{n(n-1)(n-k+1)}{n^k} \lambda^k \left(1-\frac{\lambda}{n}\right)^{n-k} \\ &= \frac{\lambda^k}{k!} e^{-\lambda}, \end{aligned} \tag{1.88}$$

where in the last step we utilize the fact that k is finite. The parameter λ represents the average number of events occurring in a fixed interval of time or space, which can be seen from the expectation and variance

$$\mathbb{E}(X) = np = \lambda, \tag{1.89a}$$

$$\text{Var}(X) = np(1-p) = \lambda(1-\frac{\lambda}{n}) = \lambda. \tag{1.89b}$$

The generating function of Poisson distribution reads

$$G_X(s) = \lim_{n\to\infty} \left(\frac{\lambda}{n}s + 1 - \frac{\lambda}{n}\right)^n = e^{\lambda(s-1)}, \tag{1.90}$$

which gives the probability distribution as

$$f_X(k) = \frac{1}{k!} \frac{\mathrm{d}^k}{\mathrm{d}s^k} G_X(s)\bigg|_{s=0} = \frac{\lambda^k}{k!} e^{-\lambda}.$$

As the possible number of events goes to infinity, the Poisson distribution indeed means that the events occur independently at a constant average rate, and we are indeed calculating the probability of events happening rate, say in a unit time interval.

Sometimes, we want to know the distribution of how many times do we need to try before certain number of events happens. This is known as the **negative-binomial**

distribution $X = nb(r, p)$, where we have

$$f_X(k \geq r) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \tag{1.91}$$

where the selective coefficient means that we need $r-1$ times of events happened before the r-th event occurs. The distribution is normalized by

$$\begin{aligned}\sum_{k=r}^{\infty} f_X(k) &= \sum_{k=r}^{\infty} \binom{k-1}{r-1} p^r (1-p)^{k-r} = p^r \sum_{n=0}^{\infty} \binom{n+r-1}{r-1} (1-p)^n \\ &= p^r (1-(1-p))^{-r} = 1.\end{aligned}$$

where we have utilized the generalized binomial theorem as a Taylor series

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \cdots = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k,$$

and

$$\begin{aligned}\binom{-r}{k} &= \frac{-r(-r-1)\cdots(-r-k+1)}{k!} = (-1)^k \frac{(r+k-1)(r+k-2)\cdots r}{k!} \\ &= (-1)^k \binom{r+k-1}{k} = (-1)^k \binom{r+k-1}{r-1}.\end{aligned}$$

The expectation and variance of the negative-binomial distribution are

$$\mathbb{E}(X) = \frac{r(1-p)}{p}, \tag{1.92a}$$

$$\mathrm{Var}(X) = \frac{r(1-p)}{p^2}. \tag{1.92b}$$

This is obtained by

$$\begin{aligned}\mathbb{E}(X) &= \sum_{k=r}^{\infty} k \binom{k-1}{r-1} p^r (1-p)^{k-r} = \sum_{k=r}^{\infty} \frac{k!}{(r-1)!(k-r)!} p^r (1-p)^{k-r} \\ &= p^r \sum_{n=0}^{\infty} \frac{(n+r)!}{(r-1)!n!} (1-p)^n = r p^r \sum_{n=0}^{\infty} \frac{(n+r)!}{r!n!} (1-p)^n \\ &= r p^r \sum_{n=0}^{\infty} \binom{-(r+1)}{n} (1-p)^n \\ &= r p^r p^{-(r+1)} = \frac{r}{p}, \\ \mathrm{Var}(X) &= \mathbb{E}(X^2) - (\mathbb{E}(X))^2 \\ &= \sum_{k=r}^{\infty} k^2 \binom{k-1}{r-1} p^r (1-p)^{k-r} - \frac{r^2}{p^2} \\ &= \sum_{k=r}^{\infty} \frac{(k+1)! - k!}{(r-1)!(k-r)!} p^r (1-p)^{k-r} - \frac{r^2}{p^2} \\ &= r(r+1) p^r p^{-(r+2)} - \frac{r}{p} - \frac{r^2}{p^2} \\ &= \frac{r(1-p)}{p^2}.\end{aligned}$$

The generating function of the negative-binomial distribution is

$$G_X(s) = \left(\frac{p}{1-(1-p)s}\right)^r. \tag{1.93}$$

The derivation of which is left after geometric distribution.

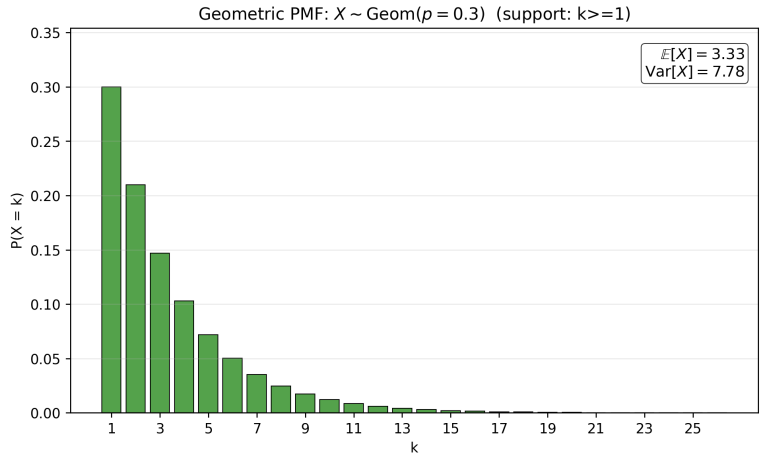


Figure 1.4: Probability mass function of a geometric distribution.

When $r = 1$, the negative-binomial distribution degrades into the **geometric distribution**

$$f_X(k) = (1-p)^{k-1}p, \tag{1.94}$$

for $k = 1, 2, \ldots$, where p is the probability of success in each trial. Then following from the negative-binomial distribution, we have expectation and variance reads

$$\mathbb{E}(X) = \frac{1}{p}, \tag{1.95a}$$

$$\text{Var}(X) = \frac{1-p}{p^2}. \tag{1.95b}$$

One indeed sees that the negative-binomial distribution can be treated as r summation of independent geometric random variables. This is reasonable, since on the ground, each event is an independent Bernoulli event, and whether next trial gives positive result is independent of the history. We thus derive the generating function as

$$\begin{aligned} G_X(s) = \mathbb{E}(s^X) &= \sum_{k=1}^{\infty} s^k (1-p)^{k-1} p \\ &= ps \sum_{k=1}^{\infty} \left((1-p)s\right)^{k-1} = \frac{ps}{1-(1-p)s}. \end{aligned} \tag{1.96}$$

The generating function of negative-binomial distribution is the r-th power of the generating function of the geometric distribution.

Combining the above distributions can lead us to the famous randomly stopped machine problem: assuming in each day, the production machine has probability p to stop, making the continuous working time Y obey geometric distribution $Y = \text{Geo}(p)$. The number of outcome products X follows a Poisson distribution conditioned by the working

time $f_{X|Y} = \text{Poisson}(\mu Y)$. Then to obtain the expectation and variance of X, we firstly have

$$\mathbb{E}(X|Y) = \text{Var}(X|Y) = \mu Y,$$

and then calculate

$$\begin{aligned}
\mathbb{E}(X) &= \mathbb{E}_Y(\mathbb{E}(X|Y)) = \mu \mathbb{E}_Y(Y) = \frac{\mu}{p}, \\
\text{Var}(X) &= \mathbb{E}_Y(\text{Var}(X|Y)) + \text{Var}_Y(\mathbb{E}(X|Y)) \\
&= \mu \mathbb{E}_Y(Y) + \mu^2 \text{Var}_Y(Y) = \frac{\mu}{p} + \frac{\mu^2(1-p)}{p^2}.
\end{aligned}$$

1.2.2 Continuous variables

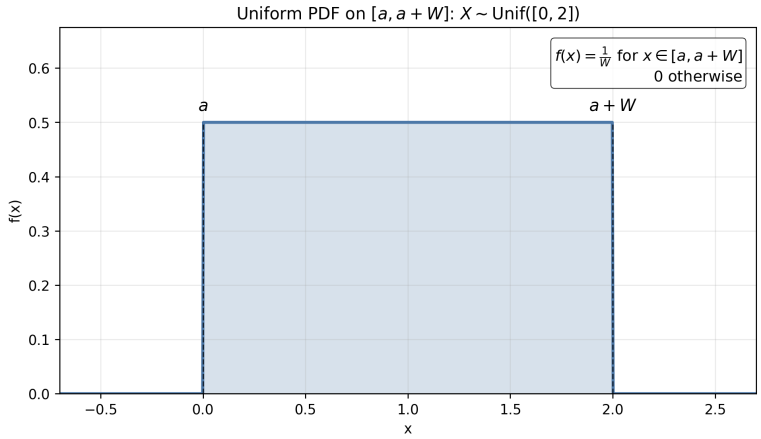


Figure 1.5: Probability density function of a uniform distribution.

Let us write again the **uniform distribution** as

$$f_X(x) = \frac{1}{W}, \ x \in [a, a+W], \tag{1.97}$$

and zero otherwise. The expectation and variance reads

$$\mathbb{E}(X) = a + \frac{W}{2}, \tag{1.98a}$$

$$\text{Var}(X) = \frac{W^2}{12}. \tag{1.98b}$$

For continuous distributions, it is possible that $\mathbb{E}(s^X)$ does not exist, and one does not obtain the discrete probability distribution. One can, however, uses the momentum gen-

erating function

$$\begin{aligned} M_X(t) = \mathbb{E}(e^{tX}) &= \int_a^{a+W} e^{tx} \frac{1}{W} \mathrm{d}x \\ &= \frac{1}{Wt}\left(e^{t(a+W)} - e^{ta}\right) = \frac{1}{W}\sum_{n=0}^{\infty} \frac{t^{n-1}}{n!}\left((a+W)^n - a^n\right) \\ &= \frac{1}{W}\sum_{n=1}^{\infty} \frac{t^{n-1}}{n!}\left((a+W)^n - a^n\right) = \frac{1}{W}\sum_{n=0}^{\infty} \frac{t^n}{(n+1)!}\left((a+W)^{n+1} - a^{n+1}\right) \\ &= \frac{1}{W}\sum_{n=0}^{\infty} \frac{t^n}{n!}\frac{(a+W)^{n+1} - a^{n+1}}{n+1}. \end{aligned} \tag{1.99}$$

and the moments of orders read

$$\mathbb{E}(X^n) = M_X^{(n)}(0) = \frac{(a+W)^{n+1} - a^{n+1}}{W(n+1)}. \tag{1.100}$$

For instance, we have $\mathbb{E}(X) = a + \frac{W}{2}$ and $\mathrm{Var}(X) = \frac{W^2}{12}$.

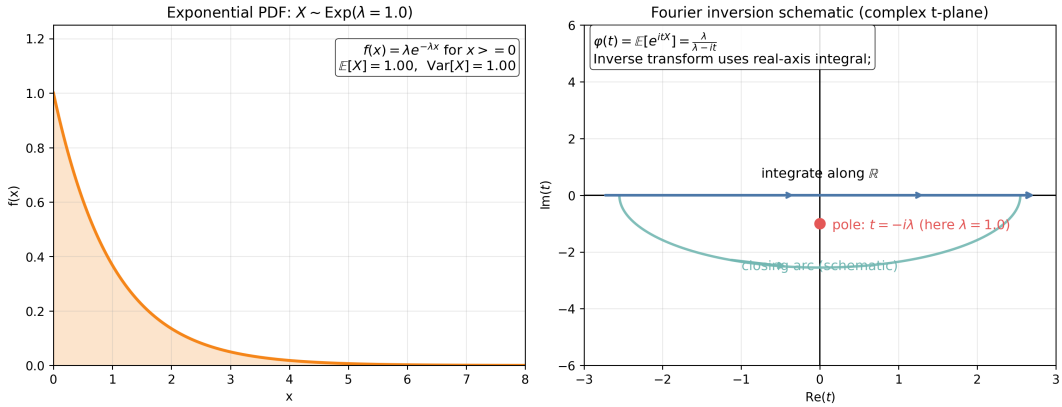


Figure 1.6: (a) Probability density function of an exponential distribution. (b) Schematic diagram for the contour integral of the characteristic function.

In Poisson distribution, we account for the number of events occurring in a fixed interval of time or space, where the events occur independently at a constant average rate. Nevertheless, the complementary situation is also important, where people need to know how long they will wait between two sequential events, and that comes to the **exponential distribution**. Denoting λ as the rate that events happen, then by assumption of independent events, at time interval $[t, t+\Delta t]$, the probability of number of events to happen obey the Poisson distribution

$$f_X(k, t, t+\Delta t) = \frac{(\lambda \Delta t)^k e^{-\lambda \Delta t}}{k!},$$

which is independent of t and can be written as $f_X(k, \Delta t)$. As $\Delta t \to 0$, the distribution degrades into nearly Bernoulli distribution, that the possibility with one event happening is $f_X(1, \Delta) = \lambda \Delta t$, and with more events to be $O((\Delta t)^2)$. The probability of zero event is $f_X(0, \Delta t) = 1 - \lambda \Delta t + O((\Delta t)^2)$. Ignoring the high order terms, and by assumption

of independent events, we can write down the probability that no events happen in time interval T as

$$f_X(0,T) = (f_X(0,T/N))^N = \left(1 - \frac{\lambda T}{N}\right)^N \overset{N\to\infty}{\to} e^{-\lambda T}. \tag{1.101}$$

This can also be understood as the probability that no events happen in time interval T, as $P(t \geq T)$, and by derivative, we obtain the PDF of exponential distribution

$$f_X(T) = \frac{\mathrm{d}}{\mathrm{d}T} P(t \leq T) = \frac{\mathrm{d}}{\mathrm{d}T}(1 - P(t \geq T)) = \lambda e^{-\lambda T}. \tag{1.102}$$

Or in common symbol using x as variable,

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0. \tag{1.103}$$

The expectation and variance of exponential distribution are easy to obtain

$$\mathbb{E}(X) = \frac{1}{\lambda}, \tag{1.104a}$$

$$\mathrm{Var}(X) = \frac{1}{\lambda^2}, \tag{1.104b}$$

which are also easy to understand, as the average waiting time should be inverse of happening rate. The standard deviation equals the average, reflecting the property of memoryless. The moment generating function is

$$M_X(t) = \mathbb{E}(e^{tX}) = \int_0^\infty e^{tx} \lambda e^{-\lambda x} \mathrm{d}x = \frac{\lambda}{\lambda - t}, \quad t < \lambda, \tag{1.105}$$

which gives moments of all orders by

$$\mathbb{E}(X^n) = M_X^{(n)}(0) = \frac{n!}{\lambda^n}, \quad n = 1, 2, \ldots \tag{1.106}$$

The characteristic function of exponential distribution reads

$$\varphi(t) = M_X(it) = \frac{\lambda}{\lambda - it}, \quad t \in \mathbb{R}, \tag{1.107}$$

and we can restore the PDF of exponential distribution by

$$\begin{aligned} f_X(x) &= \frac{1}{2\pi} \int_{-\infty}^\infty \frac{\lambda}{\lambda - it} e^{-itx} \mathrm{d}t \\ &= -\frac{1}{2\pi} 2\pi i \frac{\lambda}{-i} e^{-i(-i\lambda x)} = \lambda e^{-\lambda x}, \end{aligned} \tag{1.108}$$

where the contour integral follows diagram in Fig. **??**(b), which contains in inverse sign by opposite integral contour.

Claim 1. *The exponential distribution is also the distribution that maximizes the Shannon entropy for a normalized PDF with fixed mean.*

Proof. Let $\mathbb{E}(X) = \mu$, $\int_0^\infty \mathrm{d}x\ f(x) = 1$, then the Shannon entropy of X reads

$$S_X = -\int_0^\infty \mathrm{d}x\ f(x) \ln f(x). \tag{1.109}$$

Now we need to maximize S_X under the above two constraints, for which we construct the corresponding functional

$$S(f) = S_X + \lambda_0 \left(\int_0^\infty \mathrm{d}x \; f(x) - 1 \right) + \lambda_1 \left(\int_0^\infty \mathrm{d}x \; x f(x) - \mu \right) = \int \mathrm{d}x \; \mathscr{L}(f), \quad (1.110)$$

where the Lagrangian reads

$$\mathscr{L} = -f(x) \ln f(x) + \lambda_0 f(x) + \lambda_1 x f(x). \quad (1.111)$$

The corresponding Euler-Lagrange equation reads

$$0 = \frac{\delta \mathscr{L}}{\delta f} = -\ln f(x) - 1 + \lambda_0 + \lambda_1 x, \quad (1.112)$$

which leads to

$$f(x) = e^{\lambda_0 - 1 + \lambda_1 x}.$$

By normalization and mean, we obtain that

$$e^{\lambda_0 - 1} = \frac{1}{\mu}, \quad \lambda_1 = -\frac{1}{\mu}.$$

Substituting these values back into the expression for $f(x)$, we get

$$f(x) = \frac{1}{\mu} e^{-\frac{x}{\mu}}.$$

This is the PDF of an exponential distribution with rate parameter $\lambda = \frac{1}{\mu}$. □

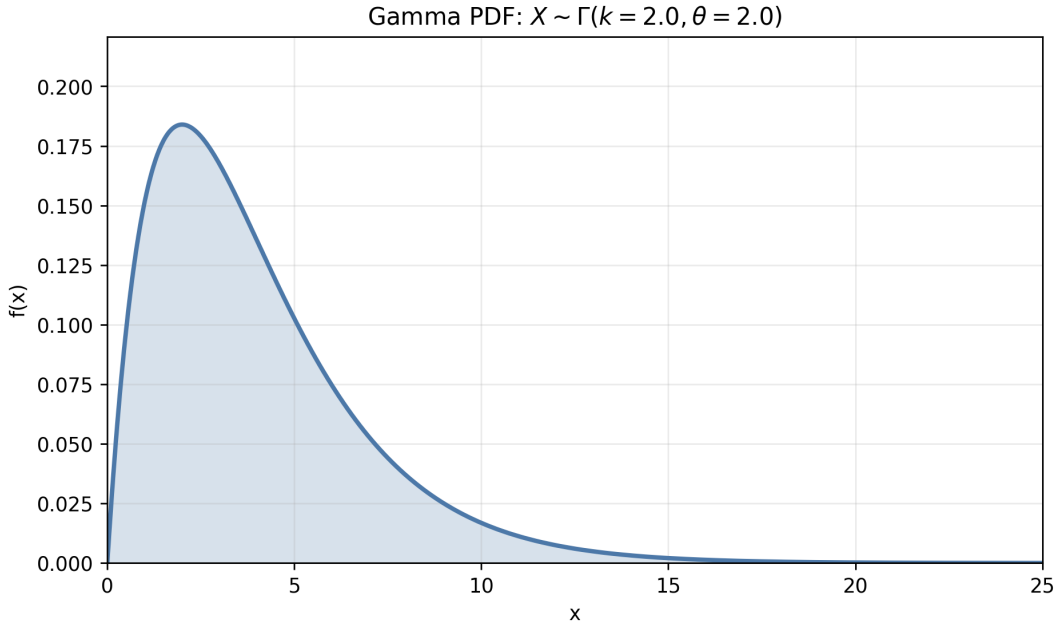


Figure 1.7: Probability density function of a gamma distribution.

If one has n independent exponential variables $\{Y_i\}_{i=1}^n$, then their sum $X = \sum_{i=1}^n Y_i$ follows a **gamma distribution** with shape parameter n and rate parameter λ. To derive its PDF, we may firstly write down its characteristic function

$$\varphi(t) = \left(\frac{\lambda}{\lambda - it} \right)^n, \quad t \in \mathbb{R}, \quad (1.113)$$

and

$$\begin{aligned} f_X(x) &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\left(\frac{\lambda}{\lambda - it}\right)^n e^{-itx}\mathrm{d}t \\ &= -\frac{1}{2\pi}\frac{2\pi i}{(n-1)!}\left(\frac{\lambda}{-i}\right)^n (-ix)^{n-1}e^{-ix(-i\lambda)} \\ &= \frac{\lambda^n x^{n-1}e^{-\lambda x}}{(n-1)!} = \frac{\lambda^n}{\Gamma(n)}x^{n-1}e^{-\lambda x}, \quad x \geq 0. \end{aligned} \tag{1.114}$$

The Gamma function is defined as

$$\Gamma(z) = \int_0^{\infty} t^{z-1}e^{-t}\mathrm{d}t, \quad \mathrm{Re}(z) > 0, \tag{1.115}$$

from which we see that the gamma distribution is automatically normalized. From the MGF

$$M_X(t) = \left(\frac{\lambda}{\lambda - t}\right)^n, \tag{1.116}$$

we see that

$$\mathbb{E}(X^m) = M_X^{(m)}(0) = \frac{n(n+1)\cdots(n+m-1)}{\lambda^m} = \frac{\Gamma(m+n)}{\Gamma(n)\lambda^m}, \tag{1.117}$$

with basically expectation and variance

$$\mathbb{E}(X) = \frac{n}{\lambda}, \tag{1.118a}$$

$$\mathrm{Var}(X) = \frac{n}{\lambda^2}, \tag{1.118b}$$

as n times of that in exponential distribution. The gamma distribution can also be obtained by involution and deduction. Say we start from $n = 2$,

$$\begin{aligned} f_{X_{(2)}}(x) &= \int_{y_1+y_2=x} \mathrm{d}y_1\mathrm{d}y_2 \; f_{Y_1,Y_2}(y_1, y_2) \\ &= \int_0^x \mathrm{d}y_1 \; f_{Y_1}(y_1)f_{Y_2}(x - y_1) \\ &= \lambda^2 e^{-\lambda x}\int_0^x \mathrm{d}y_1 = \lambda^2 x e^{-\lambda x}, \end{aligned}$$

and by assuming

$$f_{X_{(n)}}(x) = \mathrm{Gamma}(n, \lambda) = \frac{\lambda^n}{\Gamma(n)}x^{n-1}e^{-\lambda x}, \tag{1.119}$$

we can perform calculation

$$\begin{aligned} f_{X_{(n)}}(x) &= \int_0^x \mathrm{d}y \; f_{X_{(n-1)}}(y)f_{Y_1}(x - y) \\ &= \frac{\lambda^{n-1}}{\Gamma(n-1)}\lambda\int_0^x \mathrm{d}y \; y^{n-2}e^{-\lambda y}e^{-\lambda(x-y)} \\ &= \frac{\lambda^n}{\Gamma(n)}x^{n-1}e^{-\lambda x}, \end{aligned}$$

as a verification. We may extend the number n to arbitrary positive number.

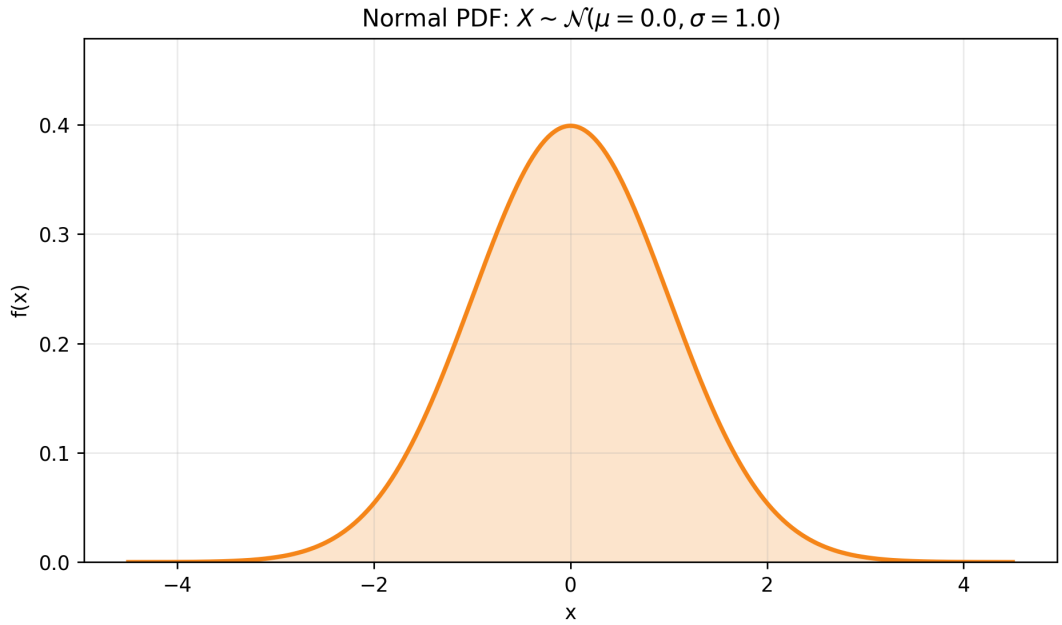


Figure 1.8: Probability density function of a normal distribution.

We have seen that the exponential distribution can be deduced by constraints of normalization and mean, and now if we add further constraint over the variance $\mathrm{Var}(X) = \sigma^2$, or specifically, we need

$$\int_{-\infty}^{\infty} \mathrm{d}x\ f_X(x) = 1,\ \int_{-\infty}^{\infty} \mathrm{d}x\ x f_X(x) = \mu,\ \int_{-\infty}^{\infty} \mathrm{d}x\ x^2 f_X(x) = \mu^2 + \sigma^2, \tag{1.120}$$

we obtain the famous Gauss distribution, aka the **normal distribution**

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}.$$

Proof. Similarly, we construct the action

$$\begin{aligned} S(f) = &-\int_{-\infty}^{\infty} \mathrm{d}x\ f(x) \ln f(x) + \lambda_0 \left(\int_{-\infty}^{\infty} \mathrm{d}x\ f(x) - 1 \right) \\ &+ \lambda_1 \left(\int_{-\infty}^{\infty} \mathrm{d}x\ x f(x) - \mu \right) + \lambda_2 \left(\int_{-\infty}^{\infty} \mathrm{d}x\ x^2 f(x) - (\mu^2 + \sigma^2) \right), \end{aligned} \tag{1.121}$$

with Lagrangian

$$\mathscr{L} = -f(x) \ln f(x) + \lambda_0 f(x) + \lambda_1 x f(x) + \lambda_2 x^2 f(x), \tag{1.122}$$

and the Euler-Lagrange equation reads

$$0 = \frac{\delta \mathscr{L}}{\delta f} = -\ln f(x) - 1 + \lambda_0 + \lambda_1 x + \lambda_2 x^2, \tag{1.123}$$

which implies

$$f(x) = e^{-1+\lambda_0+\lambda_1 x+\lambda_2 x^2}. \tag{1.124}$$

By normalizing the density and matching the constraints, we obtain the normal distribution.

One indeed sees that this procedure can be generalized to distribution with more constraints, that for constraint series

$$\int \mathrm{d}x\ x^n f(x) = \mu_n,\ n = 0, 1, 2, \cdots, N,\ \mu_0 = 1, \tag{1.125}$$

we have Lagrangian

$$\mathscr{L} = -f(x)\ln f(x) + \sum_{n=0}^{N} \lambda_n x^n f(x), \tag{1.126}$$

we have EoM

$$-\ln f(x) - 1 + \sum_{n=0}^{N} \lambda_n x^n = 0, \tag{1.127}$$

which implies

$$f(x) = e^{-1+\sum_{n=0}^{N}\lambda_n x^n}. \tag{1.128}$$

□

The Gauss distribution is thus the most random and unbiased distribution under finite information about mean and variance, which gives basis for the parameter estimation to be discussed later. One is able to show that all $n \geq 3$ cumulants equal to zero for Gauss distribution, as the MGF reads

$$\begin{aligned} M_X(t) = \mathbb{E}(e^{tX}) &= \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} \mathrm{d}x\ e^{tx} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \\ &= \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} \mathrm{d}x\ \exp\left(-\frac{1}{2}\left(\frac{x-(\mu t+\sigma^2)}{\sigma}\right)^2 + \mu t + \frac{t^2\sigma^2}{2}\right) \\ &= \exp\left(\mu t + \frac{t^2\sigma^2}{2}\right). \end{aligned} \tag{1.129}$$

We thus have

$$K_X(t) = \ln M_X(t) = \mu t + \frac{\sigma^2 t^2}{2}, \tag{1.130}$$

with 1-st and 2-nd order of cumulants being mean and variance, while higher-order cumulant vanishes, thus a Gauss distribution is fully determined by its mean and variance, and is usually signalized as $N(\mu, \sigma)$. By the linearity of mean and variance, we have that the new variable $\tilde{X} = \frac{X-\mu}{\sigma}$ has mean 0 and variance 1, thus obeying the so-called standard normal distribution $N(0, 1)$

$$f_{\tilde{X}}(\tilde{x}) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\tilde{x}^2}. \tag{1.131}$$

Theorem 1.2.1. *(Central Limit Theorem) Assuming $\{X_i\}_{i=1}^n$ are independent and identically distributed random variables with finite mean μ and variance σ^2, then*

$$\frac{1}{\sqrt{n}\sigma}\sum_{i=1}^{n}(X_i - \mu) \to N(0, 1) \quad \textit{as } n \to \infty. \tag{1.132}$$

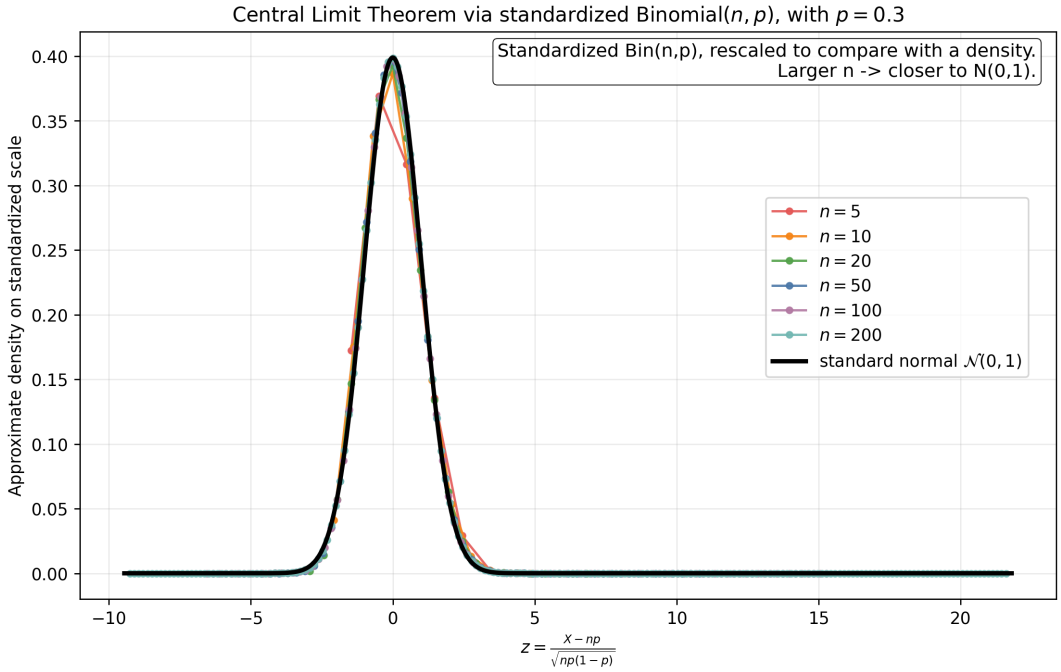


Figure 1.9: An illustration for the convergence of binomial distribution to normal distribution as n increases.

Before proving, let us firstly look at an example, that the binomial distribution $X = b(n,p)$, and our claim is that $X \to N(np, \sqrt{np(1-p)})$ as $n \to \infty$. To see this, we let $Z = \frac{X-np}{\sqrt{np(1-p)}}$, which should goes to $N(0,1)$. We prove this by showing their MGF or CGF are the same, meaning that we calculate

$$\begin{aligned} M_Z(t) = E_Z(e^{tZ}) &= e^{-\frac{npt}{\sqrt{np(1-p)}}} \left(1 - p + pe^{t/\sqrt{np(1-p)}}\right)^n \\ &= e^{-\frac{npt}{\sqrt{np(1-p)}}} \left(1 - p + p\left(1 + \frac{t}{\sqrt{np(1-p)}} + \frac{t^2}{2(np(1-p))} + \cdots\right)\right)^n \\ &= e^{-npu}\left(1 + pu + \frac{1}{2}pu^2 + O(u^3)\right)^n, \end{aligned}$$

where $u = \frac{t}{\sqrt{np(1-p)}}$. Notice that the probability of discrete variable does not change by shifting and scaling. Then we calculate

$$\begin{aligned} K_Z(t) = \ln M_Z(t) &= -npu + n\ln\left(1 + pu + \frac{1}{2}pu^2 + O(u^3)\right) \\ &= -npu + n(pu + \frac{1}{2}pu^2 + O(u^3)) - n\frac{p^2u^2}{2} + O(u^3) \\ &= \frac{1}{2}np(1-p)u^2 = \frac{t^2}{2}, \end{aligned}$$

which is exactly that of $N(0,1)$. Note that this requires p to be intermediate, i.e., $0 < p < 1$.

Proof. The proof is similar to the binomial case. That we firstly write $Y_i = \frac{X_i - \mu}{\sigma}$, which has mean 0 and variance 1. Then we can express the target variable as $Z = \frac{1}{\sqrt{n}}\sum_{i=1}^n Y_i$.

Further assume that we have MGF of Y_i as $M_{Y_i}(t) = \mathbb{E}_{Y_i}(e^{tY_i})$, which is the same for all i. By independence of Y_i, we have

$$\begin{aligned} M_Z(t) = \mathbb{E}_Z(e^{tZ}) &= \mathbb{E}_Z\left(\exp\left(\frac{t}{\sqrt{n}}\sum_{i=1}^{n} Y_i\right)\right) \\ &= \prod_{i=1}^{n} \mathbb{E}_{Y_i}\left(\exp\left(\frac{t}{\sqrt{n}}Y_i\right)\right) = \prod_{i=1}^{n} M_{Y_i}\left(\frac{t}{\sqrt{n}}\right) = \left[M_Y\left(\frac{t}{\sqrt{n}}\right)\right]^n. \end{aligned} \tag{1.133}$$

Now we have that

$$K_Z(t) = \ln M_Z(t) = n \ln M_Y\left(\frac{t}{\sqrt{n}}\right). \tag{1.134}$$

However, notice that by the suppression of factor $\frac{1}{\sqrt{n}}$, we can expand the MGF $M_Z(t)$ around $t = 0$ using mean and variance as (see Eq. **??**)

$$M_Y\left(\frac{t}{\sqrt{n}}\right) = 1 + \frac{t^2}{2n} + O(\frac{1}{n^{3/2}}),$$

and

$$K_Y\left(\frac{t}{\sqrt{n}}\right) = \ln M_Y\left(\frac{t}{\sqrt{n}}\right) = \frac{t^2}{2n} + O(\frac{1}{n^{3/2}}).$$

Thus, we have

$$K_Z(t) = \frac{t^2}{2} + O(\frac{1}{n^{1/2}}), \tag{1.135}$$

which is just the standard CGF of $N(0, 1)$. □

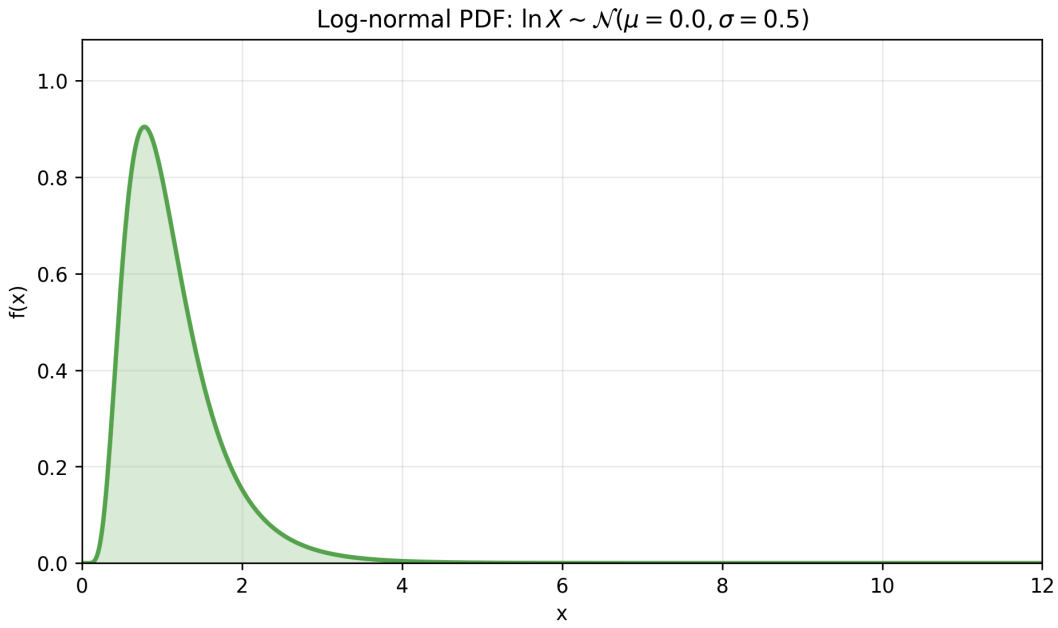


Figure 1.10: Probability density function of a log-normal distribution.

Sometimes, we meet with the **log-normal distribution**, where $X = \ln Y$ with $X = N(\mu, \sigma^2)$. By invariance of measure

$$f_Y(y)\mathrm{d}y = f_X(x)\mathrm{d}x,$$

it is easy to find that

$$f_Y(y) = \frac{1}{y\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{\ln y - \mu}{\sigma}\right)^2\right), \quad y > 0. \tag{1.136}$$

One can show that the expectation and variance of log-normal distribution read

$$\mathbb{E}(X) = e^{\mu+\frac{1}{2}\sigma^2}, \tag{1.137a}$$

$$\mathrm{Var}(X) = (e^{\sigma^2} - 1)e^{2\mu+\sigma^2}. \tag{1.137b}$$

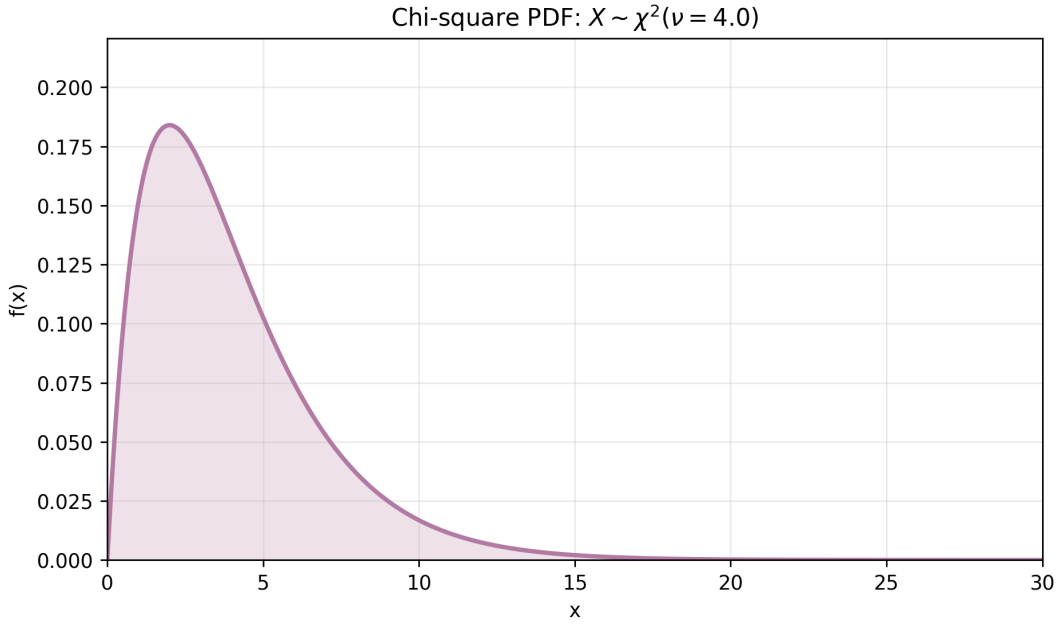


Figure 1.11: Probability density function of a χ-square distribution.

In inferential statistics, at many times we will meet with the so-called **chi-squared distribution**, that by independent standard normal distribution $\{Z_i = N(0,1)\}_{i=1}^n$, we have

$$X = \sum_{i=1}^{n} Z_i^2 = \chi^2(n). \tag{1.138}$$

To study its property, we firstly focus on the case with $n = 1$, that is, $X_1 = Z^2$ with $Z = N(0,1)$. By invariance of measure, we have

$$f_{X_1}(x)\mathrm{d}x = \sum_{z^2=x} f_Z(z)\mathrm{d}z = \frac{1}{\sqrt{2\pi}}\left(e^{-\frac{1}{2}\sqrt{x}^2} + e^{-\frac{1}{2}(-\sqrt{x})^2}\right)\frac{\mathrm{d}x}{2\sqrt{x}} = \frac{1}{\sqrt{2\pi x}}e^{-\frac{x}{2}}\mathrm{d}x,$$

i.e.,

$$f_{X_1}(x) = \frac{1}{\sqrt{2\pi x}}e^{-\frac{x}{2}}, \quad x > 0. \tag{1.139}$$

We thus obtain the MGF of $\chi^2(1)$ as

$$M_{X_1}(t) = \mathbb{E}(e^{tX_1}) = \int_0^\infty e^{tx}\frac{1}{\sqrt{2\pi x}}e^{-\frac{x}{2}}\mathrm{d}x = (1-2t)^{-\frac{1}{2}}, \quad t < \frac{1}{2}, \tag{1.140}$$

by a simple change of variable $u = \sqrt{x}$. From the MGF, we can easily obtain the expectation and variance of $\chi^2(1)$ as

$$\mathbb{E}(X_1) = \frac{\mathrm{d}}{\mathrm{d}t} M_{X_1}(t)\bigg|_{t=0} = 1, \tag{1.141a}$$

$$\mathrm{Var}(X_1) = \frac{\mathrm{d}^2}{\mathrm{d}t^2} M_{X_1}(t)\bigg|_{t=0} - \left(\frac{\mathrm{d}}{\mathrm{d}t} M_{X_1}(t)\bigg|_{t=0}\right)^2 = 2. \tag{1.141b}$$

For general n, by independence, we can write down the MGF of $\chi^2(n)$ as

$$M_X(t) = \mathbb{E}(e^{tX}) = \prod_{i=1}^{n} \mathbb{E}(e^{tZ_i^2}) = (1-2t)^{-\frac{n}{2}}, \quad t < \frac{1}{2}, \tag{1.142}$$

which gives expectation and variance as

$$\mathbb{E}(X) = n, \tag{1.143a}$$

$$\mathrm{Var}(X) = 2n, \tag{1.143b}$$

which can also be obtained by the linearity of mean and variance. To obtain the PDF, we can perform the inverse Fourier transform of the characteristic function $\varphi(t) = M_X(it) = (1-2it)^{-\frac{n}{2}}$ as

$$\begin{aligned} f_X(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} (1-2it)^{-\frac{n}{2}} e^{-itx} \mathrm{d}t \\ &= -\frac{1}{2\pi} \frac{2\pi i}{(n/2-1)!} \left(\frac{1}{-i}\right)^{\frac{n}{2}} (-ix)^{\frac{n}{2}-1} e^{-ix(-i/2)} \\ &= \frac{1}{2^{\frac{n}{2}}\Gamma(n/2)} x^{\frac{n}{2}-1} e^{-\frac{x}{2}}, \quad x > 0, \end{aligned} \tag{1.144}$$

which is a specific case of gamma distribution with shape parameter $\frac{n}{2}$ and rate parameter $\frac{1}{2}$, i.e., $\chi^2(n) = \mathrm{Gamma}(\frac{n}{2}, \frac{1}{2})$.

In area of inferential statistics, the chi-squared distribution is widely used for hypothesis testing and confidence interval estimation, especially in the context of variance and goodness-of-fit tests. At many times, we will meet with the so-called ***t*-distribution**, that by independent standard normal distribution $Z = N(0,1)$ and $\chi^2(n)$ distribution $Y = \chi^2(n)$, we have

$$X = \frac{Z}{\sqrt{Y/n}} = t(n). \tag{1.145}$$

To study its property, by independence, we firstly write down the joint PDF of Z and Y as

$$f_{Z,Y}(z,y) = f_Z(z) f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \frac{1}{2^{\frac{n}{2}}\Gamma(n/2)} y^{\frac{n}{2}-1} e^{-\frac{y}{2}}. \tag{1.146}$$

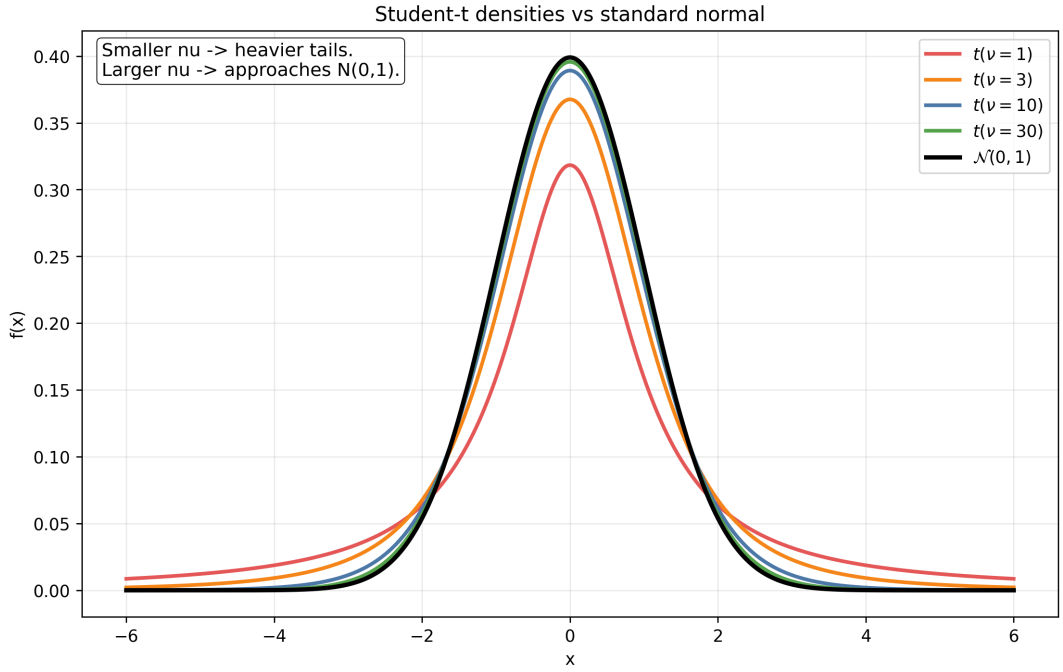


Figure 1.12: Probability density function of a t-distribution.

By invariance of measure, we have

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} \mathrm{d}z \int_0^{\infty} \mathrm{d}y \; f_{Z,Y}(z,y)\delta\left(x - \frac{z}{\sqrt{y/n}}\right) \\ &= \int_0^{\infty} \mathrm{d}y \; \sqrt{\frac{y}{n}} \frac{1}{\sqrt{2\pi}} e^{-\frac{yx^2}{2n}} \frac{1}{2^{\frac{n}{2}}\Gamma(n/2)} y^{\frac{n}{2}-1} e^{-\frac{y}{2}} \\ &= \frac{1}{\sqrt{2n\pi}2^{\frac{n}{2}}\Gamma(n/2)} \int_0^{\infty} \mathrm{d}y \; y^{\frac{n+1}{2}-1} e^{-\frac{y}{2}\left(1+\frac{x^2}{n}\right)} \\ &= \frac{1}{\sqrt{2n\pi}2^{\frac{n}{2}}\Gamma(n/2)} \frac{\Gamma(\frac{n+1}{2})}{\left(\frac{1}{2}\left(1+\frac{x^2}{n}\right)\right)^{\frac{n+1}{2}}} \\ &= \frac{1}{\sqrt{n\pi}} \frac{\Gamma(\frac{n+1}{2})}{\Gamma(\frac{n}{2})} \left(1+\frac{x^2}{n}\right)^{-\frac{n+1}{2}}, \end{aligned}$$

where from the second equality, we utilized the integral of δ-function

$$\int \mathrm{d}x \; \delta(x - f(x))g(x) = \sum_{x_0} \frac{g(x_0)}{|f'(x_0)|},$$

while in the fourth equality, we utilized a variant of Gamma function,

$$\int_0^{\infty} u^{\alpha-1} e^{-\beta u} \mathrm{d}u = \frac{\Gamma(\alpha)}{\beta^{\alpha}}.$$

This gives the PDF of t-distribution as

$$f_X(x) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}\Gamma\left(\frac{n}{2}\right)} \left(1+\frac{x^2}{n}\right)^{-\frac{n+1}{2}}, \quad x \in \mathbb{R}. \tag{1.147}$$

If the above derivation is not satisfactory enough, we may also introduce $U = Y$, such that $Z = X\sqrt{U/n}$, and we have the Jacobian of transformation as

$$\left|\frac{\partial(z,y)}{\partial(x,u)}\right| = \begin{vmatrix} \partial_x z & \partial_u z \\ \partial_x y & \partial_u y \end{vmatrix} = \begin{vmatrix} \sqrt{\frac{u}{n}} & \frac{x}{2\sqrt{nu}} \\ 0 & 1 \end{vmatrix} = \sqrt{\frac{u}{n}},$$

and then from invariance of measure, we have

$$f_{X,U}(x,u)\mathrm{d}x\mathrm{d}u = f_{Z,Y}(z,y)\mathrm{d}z\mathrm{d}y,$$

we can obtain the joint PDF of X and U as

$$f_{X,U}(x,u) = f_{Z,Y}(z,y)\left|\frac{\partial(z,y)}{\partial(x,u)}\right| = f_Z(x\sqrt{u/n})f_Y(u)\sqrt{\frac{u}{n}},$$

and then by marginalization, we can obtain the PDF of t-distribution as the same as above. Notice that the t-distribution is symmetric around zero, and has heavier tails than the normal distribution, which makes it more robust to outliers in data. The power-law tail of t-distribution prevents the existence of higher-order moments, that the m-th order moment exists only when $n > m$, and thus the variance of t-distribution exists only when $n > 2$. Nevertheless, as $n \to \infty$, the t-distribution converges to the normal distribution, which can be verified by the asymptotic behavior

$$\lim_{n\to\infty}\frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}\Gamma\left(\frac{n}{2}\right)} = \frac{1}{\sqrt{2\pi}},\ \lim_{n\to\infty}\left(1+\frac{x^2}{n}\right)^{-\frac{n+1}{2}} = e^{-\frac{x^2}{2}}.$$

Chapter 2

Statistics

In the previous chapter, we have discussed the static configurations of various kinds of probability distributions. Nevertheless, these distributions do not come from nowhere, and they are constructed from our experiences, observations, and data. In this chapter, we will discuss how to estimate the parameters of a distribution from data, and how to test hypotheses about the parameters of a distribution.

2.1 Estimation and hypothesis testing

2.2 Regression analysis

Part II

Stochastic Process

Chapter 3

Branching Processes

Reproduction model.

Chapter 4

Markov Chain

Memory losing property.

Chapter 5

Stochastic Differential Equations

Continuous-time stochastic processes.

Part III

Machine Learning

Chapter 6

Multi-Layer Perceptron

Feed forward neural networks (FFNN), beginning of deep learning.

6.1 Linear regression revisited

6.2 Perceptron structure

6.3 Deep neural networks

A neural network (NN) functions as a much more powerful perceptron, or a function, a map, which contains an input, an evaluation and an output

$$\text{Input} \xrightarrow{\text{NN}} \text{Output}. \tag{6.1}$$

Bearing the name network, the structure is separated into multiple layers, and specifically,

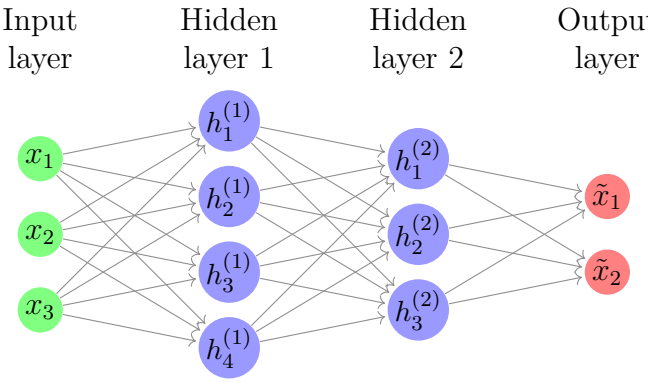


Figure 6.1: An example of multi-layer perceptron with two hidden layers.

the input, or input layer, is usually some observable data set, that is organized into a one-dimensional vector $\boldsymbol{X}_{\text{in}} = (x_1, x_2, \ldots, x_n)^T$; the evaluation consists of a series of calculating layers that transit the input through the network, also known as hidden layers $[\boldsymbol{H}^{(1)}, \boldsymbol{H}^{(2)}, \ldots, \boldsymbol{H}^{(L)}]$ with L layers, and $\boldsymbol{H}^{(l)} = (h_1^{(l)}, h_2^{(l)}, \ldots, h_{m_l}^{(l)})^T$ is the hidden vector of lth layer; finally, the output, or output layer, is usually some feature data set that aims at effectively classifying the given input dataset, and can be represented by $\boldsymbol{X}_{\text{out}} = (\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_p)^T$. One illustration diagram of the MLP deep NN is represented in Fig. **??**,

with two hidden layers for computation. Especially notice that there are already several parameters like the network depth L and number of perceptrons m_l that are adjustable in each specific network training process.

Speaking of the training process, there is one last piece of information, and possibly the most important one, that we need to fill into the NN. That denotes the weight matrix $[\mathbf{W}^{(0)}, \mathbf{W}^{(1)}, \ldots, \mathbf{W}^{(L)}]$ and the activation function Φ, with $\mathbf{W}^{(l)}$ denotes the linear map from lth layer to $(l+1)$th layer (0 for input layer) by matrix multiplication $\mathbf{W}^{(l)} \cdot \boldsymbol{H}^{(l)}$. The activation function then adds non-linearity to the evaluation process and gives the vector value of the next layer:

$$\boldsymbol{H}^{(l+1)} = \Phi(\mathbf{W}^{(l)} \cdot \boldsymbol{H}^{(l)}), \tag{6.2}$$

and in components, we notice that Φ is an element-wise action, and thus

$$h_i^{(l+1)} = \Phi\left(\sum_{j=1}^{m_l} w_{ij}^{(l)} h_j^{(l)}\right), \tag{6.3}$$

where $i = 1, 2, \ldots, m_{l+1}$ and $j = 1, 2, \ldots, m_l$. We thus have a chain of transformations from the input to the output as

$$\boldsymbol{X}_{\text{in}} \overset{\Phi(\mathbf{W}^{(0)}\cdot)}{\longrightarrow} \boldsymbol{H}^{(1)} \overset{\Phi(\mathbf{W}^{(1)}\cdot)}{\longrightarrow} \boldsymbol{H}^{(2)} \longrightarrow \cdots \longrightarrow \boldsymbol{H}^{(L)} \overset{\Phi(\mathbf{W}^{(L)}\cdot)}{\longrightarrow} \boldsymbol{X}_{\text{out}}. \tag{6.4}$$

This is not an exact sequence. As we have said, the aim of the activation function is to introduce non-linearity into the evaluation process, so that the NN can really represent some complex non-linear, or even chaotic process of realistic systems, and thus we have usually several choices in the below Table. **??**, where CDF of normal refers to the cumulative distribution function of a standard normal distribution

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2} dt.$$

ReLU refers to the rectified linear unit activation function. ELU refers to the exponential linear unit activation function.

Table 6.1: Common activation functions.

Name	**Formula**	**Characteristics**
Sigmoid	$\sigma(x) = \dfrac{1}{1+e^{-x}}$	Output in $(0,1)$, smooth, saturating
Tanh	$\tanh(x) = \dfrac{e^{x}-e^{-x}}{e^{x}+e^{-x}}$	Output in $(-1,1)$, zero-centered, saturating
ReLU	$\max(0,x)$	Non-saturating for $x>0$, sparse activations
Leaky ReLU	$\max(\alpha x, x)$ with $\alpha \in (0,1)$	Avoids dying ReLU
ELU	$\begin{cases} x & x>0 \\ \alpha(e^{x}-1) & x\le 0 \end{cases}$	Smooth negative part, mean shift
Swish	$x \cdot \sigma(x)$	Non-monotonic, smooth
GELU	$x \cdot \Phi(x)$ where Φ is CDF of normal	Smooth, used in transformers
Softmax	$\dfrac{e^{x_i}}{\sum_j e^{x_j}}$	Converts logits to probabilities (vector output)

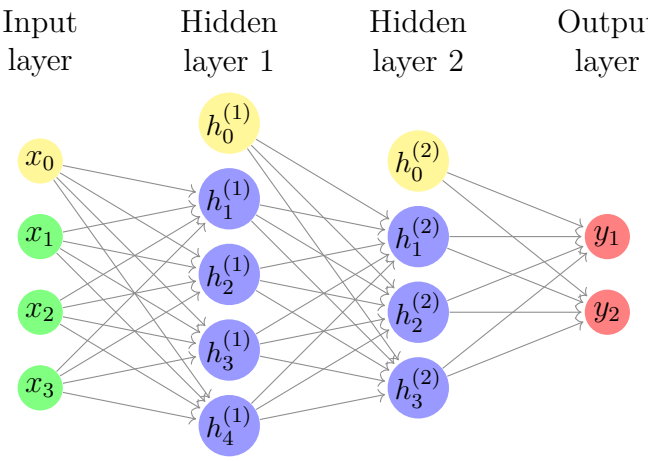


Figure 6.2: An example of multi-layer perceptron with two hidden layers and bias terms.

At many times, we need a bias node to provide a constant input to each neuron in the network, allowing the activation function to be shifted and scaled. This bias enables non-zero output even when the input is zero, and improves greatly the flexibility and expressiveness of the network. An example diagram is presented in Fig. **??**. In this case, the transit between layers is functionalized as

$$\boldsymbol{H}^{(l+1)} = \Phi\left(\mathbf{W}^{(l)}\boldsymbol{H}^{(l)} + \boldsymbol{b}^{(l)}\right), \tag{6.5}$$

where $\boldsymbol{b}^{(l)}$ is the bias vector for layer l. Notice that this looks a bit different from our diagram, but they are indeed compatible, only with a viewpoint difference that in the diagram, the bias node is added as the input of each layer, i.e., we extend $\boldsymbol{X}_{\text{in}} \to \tilde{\boldsymbol{X}}_{\text{in}} = \begin{pmatrix} x_0 \\ \boldsymbol{X}_{\text{in}} \end{pmatrix}$, $\boldsymbol{H}^{(l)} \to \tilde{\boldsymbol{H}}^{(l)} = \begin{pmatrix} h_0^{(l)} \\ \boldsymbol{H}^{(l)} \end{pmatrix}$, and enlarge the corresponding weight matrices by

$$\mathbf{W}^{(l)} \to \tilde{\mathbf{W}}^{(l)} = \begin{pmatrix} \boldsymbol{w}_0^{(l)} \\ \mathbf{W}^{(l)} \end{pmatrix} = \begin{pmatrix} w_{01}^{(l)} & w_{02}^{(l)} & \cdots & w_{0m_l}^{(l)} \\ w_{11}^{(l)} & w_{12}^{(l)} & \cdots & w_{1m_l}^{(l)} \\ \vdots & \vdots & \ddots & \vdots \\ w_{m_{l+1}1}^{(l)} & w_{m_{l+1}2}^{(l)} & \cdots & w_{m_{l+1}m_l}^{(l)} \end{pmatrix},$$

i.e., from $m_l \times m_{l+1}$ to $(m_l + 1) \times m_{l+1}$. In this case, the transit between layers reads

$$\boldsymbol{H}^{(l+1)} = \Phi\left(\tilde{\mathbf{W}}^{(l)}\tilde{\boldsymbol{H}}^{(l)}\right), \tag{6.6}$$

and each evaluated vector needs to be combined with the bias node value. The chain of transformations is expressed as

$$\begin{array}{ccccccccc} & & h_0^{(1)} & & h_0^{(2)} & & & & h_0^{(L)} \\ & & \downarrow & & \downarrow & & & & \downarrow \\ \tilde{\boldsymbol{X}}_{\text{in}} & \xrightarrow{\Phi(\tilde{\mathbf{W}}^{(0)}\cdot)} & \boldsymbol{H}^{(1)} & \xrightarrow{\Phi(\tilde{\mathbf{W}}^{(1)}\cdot)} & \boldsymbol{H}^{(2)} & \xrightarrow{\Phi(\tilde{\mathbf{W}}^{(2)}\cdot)} & \cdots & & \boldsymbol{H}^{(L)} \xrightarrow{\Phi(\tilde{\mathbf{W}}^{(L)}\cdot)} \boldsymbol{X}_{\text{out}} \end{array} \tag{6.7}$$

In this case, we can usually choose $x_0 = h_0^{(l)} = 1$, and the training only changes the enlarged weight matrices. On the other hand, we can also use the viewpoint that the

calculated bias influence represented by $\boldsymbol{b}^{(l)} = \boldsymbol{w}_0^{(l)} h_0^{(l)}$ has already been absorbed into the input of the activation function, leading to the expression in (**??**). In this case, the chain of transformation reads

$$\boldsymbol{X}_{\text{in}} \overset{\Phi(\boldsymbol{b}^{(0)}+\mathbf{W}^{(0)}\cdot)}{\longrightarrow} \boldsymbol{H}^{(1)} \overset{\Phi(\boldsymbol{b}^{(1)}+\mathbf{W}^{(1)}\cdot)}{\longrightarrow} \boldsymbol{H}^{(2)} \longrightarrow \cdots \longrightarrow \boldsymbol{H}^{(L)} \overset{\Phi(\boldsymbol{b}^{(L)}+\mathbf{W}^{(L)}\cdot)}{\longrightarrow} \boldsymbol{X}_{\text{out}}. \tag{6.8}$$

In this case, the bias vector $\boldsymbol{b}^{(l)}$, $l = 0, 1, \cdots, L$ are trainable parameters.

We have talked several times about training. What is the goal of training? It is about approaching a minimal error between the predicted outputs and the true labels, thus belonging to a minimization problem. In physics, this is analogous to finding the minimum of a free energy as function of physical configurations, and here, the function is called the loss function, or cost function. Denoting true output as $\boldsymbol{Y} = (y_1, y_2, \ldots, y_p)^T$, the loss function has general form $\mathscr{L} = \mathscr{L}(\boldsymbol{X}_{\text{out}}, \boldsymbol{Y})$, and our aim is to minimize this function w.r.t. variational parameters $w_{ij}^{(l)}$ (we have absorbed the bias terms into the weight matrices). Specifically, we have

$$\frac{\partial \mathscr{L}}{\partial w_{ij}^{(l)}} = \frac{\partial \mathscr{L}}{\partial \boldsymbol{X}_{\text{out}}} \cdot \frac{\partial \boldsymbol{X}_{\text{out}}}{\partial w_{ij}^{(l)}}, \tag{6.9}$$

with $\frac{\partial \mathscr{L}}{\partial \boldsymbol{X}_{\text{out}}}$ the derivative of loss function w.r.t. the output. For instance, given a simple mean squared error (MSE) loss function:

$$\mathscr{L} = \frac{1}{2}\|\boldsymbol{X}_{\text{out}} - \boldsymbol{Y}\|^2 = \frac{1}{2}\sum_{i=1}^{p}(\tilde{x}_i - y_i)^2, \tag{6.10}$$

we have

$$\frac{\partial \mathscr{L}}{\partial \boldsymbol{X}_{\text{out}}} = \boldsymbol{X}_{\text{out}} - \boldsymbol{Y} \tag{6.11}$$

in a vector form. Now assume that we have calculated all $\frac{\partial \mathscr{L}}{\partial w_{ij}^{(l)}}$, then a training direction can be defined by the **gradient descent** (GD) by changing parameters

$$w_{ij}^{(L)} \to \bar{w}_{ij}^{(L)} = w_{ij}^{(L)} - \eta \frac{\partial \mathscr{L}}{\partial w_{ij}^{(l)}}, \tag{6.12}$$

where $\eta > 0$ is called the **learning rate**. This holds since we can write the loss function, on the other side, as a function of variational parameters, i.e., $\mathscr{L} = \mathscr{L}(\mathbf{W}^{(l)})$, and by a Taylor expansion:

$$\begin{aligned}\mathscr{L}(\bar{\mathbf{W}}^{(l)}) &= \mathscr{L}(w_{ij}^{(L)} - \eta \frac{\partial \mathscr{L}}{\partial w_{ij}^{(l)}}) \\ &= \mathscr{L}(w_{ij}^{(L)}) - \eta \sum_{i,j} \frac{\partial \mathscr{L}}{\partial w_{ij}^{(l)}} \frac{\partial \mathscr{L}}{\partial w_{ij}^{(l)}} + \mathcal{O}(\eta^2),\end{aligned}$$

and one sees that the loss function is always directed towards a minimum.

As one can imagine, the term $\frac{\partial \boldsymbol{X}_{\text{out}}}{\partial w_{ij}^{(l)}}$ is a long chain from the output to the lth layer, and for illustration, we can utilize a single variable/scalar chain:

$$x_{\text{in}} \overset{\Phi(w^{(0)}\cdot)}{\longrightarrow} h^{(1)} \overset{\Phi(w^{(1)}\cdot)}{\longrightarrow} h^{(2)} \overset{\Phi(w^{(2)}\cdot)}{\longrightarrow} \cdots \overset{\Phi(w^{(L-1)}\cdot)}{\longrightarrow} h^{(L)} \overset{\Phi(w^{(L)}\cdot)}{\longrightarrow} x_{\text{out}}, \tag{6.13}$$

and we have

$$\begin{aligned}\frac{\partial x_{\text{out}}}{\partial w^{(l)}} &= \frac{\partial \Phi(w^{(L)}h^{(L)})}{\partial w^{(l)}} = \Phi'(w^{(L)}h^{(L)})w^{(L)}\frac{\partial h^{(L)}}{\partial w^{(l)}} \\ &= \Phi'(w^{(L)}h^{(L)})w^{(L)}\Phi'(w^{(L-1)}h^{(L-1)})w^{(L-1)}\frac{\partial h^{(L-1)}}{\partial w^{(l)}} \\ &= \cdots = \left(\prod_{k=l+1}^{L}\Phi'(w^{(k)}h^{(k)})w^{(k)}\right)\Phi'(w^{(l)}h^{(l)})h^{(l)},\end{aligned} \tag{6.14}$$

where we notice that

$$\frac{\partial(w^{(n)}h^{(n)})}{\partial w^{(l)}} = \begin{cases} 0 & \text{if } l > n, \\ h^{(n)} & \text{if } l = n, \\ w^{(n)}\frac{\partial h^{(n)}}{\partial w^{(l)}} & \text{if } l < n. \end{cases}$$

Especially, we notice the iterative relation

$$\frac{\partial x_{\text{out}}}{\partial w^{(l)}} = \frac{\partial x_{\text{out}}}{\partial w^{(l+1)}}\frac{w^{(l+1)}}{h^{(l+1)}}\Phi'(w^{(l)}h^{(l)})h^{(l)}, \tag{6.15}$$

and this gives us the idea of **back propagation** (BP), that by defining error term

$$\delta^{(l)} = \frac{\partial x_{\text{out}}}{\partial u^{(l)}},\ u^{(l)} = h^{(l)}w^{(l)}, \tag{6.16}$$

one has the error BP

$$\delta^{(l)} = \delta^{(l+1)}w^{(l+1)}\Phi'(u^{(l)}). \tag{6.17}$$

This formula can be implemented efficiently with computational algorithm. To account for the bias, we make it explicitly in the scalar chain:

$$x_{\text{in}} \overset{\Phi(b^{(0)}+w^{(0)}\cdot)}{\longrightarrow} h^{(1)} \overset{\Phi(b^{(1)}+w^{(1)}\cdot)}{\longrightarrow} h^{(2)} \longrightarrow \cdots \longrightarrow \overset{\Phi(b^{(L)}+w^{(L)}\cdot)}{\longrightarrow} x_{\text{out}}, \tag{6.18}$$

and we similarly have

$$\begin{aligned}\frac{\partial x_{\text{out}}}{\partial b^{(l)}} &= \Phi'(u^{(L)})\frac{\partial u^{(L)}}{\partial b^{(l)}} = \Phi'(u^{(L)})w^{(L)}\frac{\partial h^{(L)}}{\partial b^{(l)}} \\ &= \Phi'(u^{(L)})w^{(L)}\Phi'(u^{(L-1)})\frac{\partial u^{(L-1)}}{\partial b^{(l)}} \\ &= \cdots = \left(\prod_{k=l+1}^{L}\Phi'(u^{(k)})w^{(k)}\right)\Phi'(u^{(l)}),\end{aligned} \tag{6.19}$$

where $u^{(l)} = b^{(l)} + w^{(l)}h^{(l)}$, and we have

$$\frac{\partial(u^{(l)})}{\partial b^{(l)}} = \begin{cases} 0 & \text{if } l > n, \\ 1 & \text{if } l = n, \\ w^{(n)}\frac{\partial h^{(n)}}{\partial b^{(l)}} & \text{if } l < n. \end{cases}$$

Thus the BP of the bias terms is given by

$$\frac{\partial x_{\text{out}}}{\partial b^{(l)}} = \frac{\partial x_{\text{out}}}{\partial b^{(l+1)}}w^{(l+1)}\Phi'(u^{(l)}), \tag{6.20}$$

which is the same as the weight with $h \equiv 1$.

Now we turn back to the matrix case. For step by step, let us firstly focus on the Lth layer:

$$\frac{\partial \tilde{x}_m}{\partial w_{ij}^{(L)}} = \delta_{im} \Phi'(u_m^{(L)}) h_j^{(L)}, \tag{6.21}$$

where $\boldsymbol{U}^{(l)} = \tilde{\mathbf{W}}^{(l)} \tilde{\boldsymbol{H}}^{(l)} = \mathbf{W}^{(l)} \boldsymbol{H}^{(l)} + \boldsymbol{b}^{(l)}$. Thus, we have

$$\frac{\partial \mathscr{L}}{\partial w_{ij}^{(L)}} = \sum_m \frac{\partial \mathscr{L}}{\partial \tilde{x}_m} \frac{\partial \tilde{x}_m}{\partial w_{ij}^{(L)}} = \frac{\partial \mathscr{L}}{\partial \tilde{x}_i} \Phi'(u_i^{(L)}) h_j^{(L)},$$

or

$$\frac{\partial \mathscr{L}}{\partial \mathbf{W}^{(L)}} = \left(\frac{\partial \mathscr{L}}{\partial \boldsymbol{X}_{\text{out}}} \odot \Phi'(\boldsymbol{U}^{(L)}) \right) \cdot (\boldsymbol{H}^{(L)})^T, \tag{6.22}$$

where $\odot$ denotes the Hadamard product (element-wise product), and $\cdot$ the normal matrix product. Notice that both sides are matrices. Now for $L-1$th layer,

$$\begin{aligned}
\frac{\partial \tilde{x}_m}{\partial w_{ij}^{(L-1)}} &= \Phi'(u_m^{(L)}) \frac{\partial u_m^{(L)}}{\partial w_{ij}^{(L-1)}} = \Phi'(u_m^{(L)}) \sum_n w_{mn}^{(L)} \frac{\partial h_n^{(L)}}{\partial w_{ij}^{(L-1)}} \\
&= \Phi'(u_m^{(L)}) \sum_n w_{mn}^{(L)} \Phi'(u_n^{(L-1)}) \frac{\partial u_n^{(L-1)}}{\partial w_{ij}^{(L-1)}} \\
&= \Phi'(u_m^{(L)}) w_{mi}^{(L)} \Phi'(u_i^{(L-1)}) h_j^{(L-1)},
\end{aligned}$$

thus, we have

$$\begin{aligned}
\frac{\partial \mathscr{L}}{\partial w_{ij}^{(L-1)}} &= \sum_m \frac{\partial \mathscr{L}}{\partial \tilde{x}_m} \frac{\partial \tilde{x}_m}{\partial w_{ij}^{(L-1)}} = \sum_m \frac{\partial \mathscr{L}}{\partial \tilde{x}_m} \Phi'(u_m^{(L)}) w_{mi}^{(L)} \Phi'(u_i^{(L-1)}) h_j^{(L-1)} \\
&= \Phi'(u_i^{(L-1)}) \sum_m [(\mathbf{W}^{(L)})^T]_{im} \left(\frac{\partial \mathscr{L}}{\partial \boldsymbol{X}_{\text{out}}} \odot \Phi'(\boldsymbol{U}^{(L)}) \right)_m h_j^{(L-1)},
\end{aligned}$$

which leads to

$$\frac{\partial \mathscr{L}}{\partial \mathbf{W}^{(L-1)}} = \Phi'(\boldsymbol{U}^{(L-1)}) \odot \left[(\mathbf{W}^{(L)})^T \cdot \left(\frac{\partial \mathscr{L}}{\partial \boldsymbol{X}_{\text{out}}} \odot \Phi'(\boldsymbol{U}^{(L)}) \right) \right] \cdot (\boldsymbol{H}^{(L-1)})^T. \tag{6.23}$$

To simplify the notation, we may define

$$\boldsymbol{D}^{(l)} = \frac{\partial \mathscr{L}}{\partial \boldsymbol{X}_{\text{out}}} \odot \Phi'(\boldsymbol{U}^{(l)}), \ \boldsymbol{\delta}^{(l)} = \frac{\partial \mathscr{L}}{\partial \boldsymbol{U}^{(l)}}, \tag{6.24}$$

with the latter as the error term, and the above two equations can be written as

$$\boldsymbol{\delta}^{(L)} = \boldsymbol{D}^{(L)}, \tag{6.25a}$$

$$\boldsymbol{\delta}^{(L-1)} = \Phi'(\boldsymbol{U}^{(L-1)}) \odot \left((\mathbf{W}^{(L)})^T \cdot \boldsymbol{D}^{(L)} \right) = \Phi'(\boldsymbol{U}^{(L-1)}) \odot \left((\mathbf{W}^{(L)})^T \cdot \boldsymbol{\delta}^{(L)} \right), \tag{6.25b}$$

and this leads to our generalization as the BP of MLP:

$$\boldsymbol{\delta}^{(l)} = \Phi'(\boldsymbol{U}^{(l)}) \odot \left((\mathbf{W}^{(l+1)})^T \cdot \boldsymbol{\delta}^{(l+1)} \right), \tag{6.26}$$

for $l = L-1, L-2, \ldots, 1$, with the boundary condition $\boldsymbol{\delta}^{(L)} = \boldsymbol{D}^{(L)}$. And to get the partial derivative for weight matrix for GD, we have

$$\frac{\partial \mathscr{L}}{\partial \mathbf{W}^{(l)}} = \boldsymbol{\delta}^{(l)} \cdot (\boldsymbol{H}^{(l)})^T, \tag{6.27}$$

for $l = 1, 2, \ldots, L$. The form here is much similar to the scalar case with product rules changed. For the bias term, if we write it explicitly, we will obtain

$$\frac{\partial \mathscr{L}}{\partial \boldsymbol{b}^{(l)}} = \Phi'(\boldsymbol{U}^{(l)}) \odot \left[(\mathbf{W}^{(l+1)})^T \cdot \frac{\partial \mathscr{L}}{\partial \boldsymbol{b}^{(l+1)}} \right], \tag{6.28}$$

with boundary condition $\frac{\partial \mathscr{L}}{\partial \boldsymbol{b}^{(L)}} = \boldsymbol{D}^{(L)}$.

The training parameters are listed in the following Table **??**. Slightly changed **Nota-**

Table 6.2: MLP training parameters and backpropagation equations

Name	**Symbol**	**Dimensions**	**Formula / Description**
Input	$\mathbf{H}^{(0)} = \{\boldsymbol{X}_{\text{in}}^i\}_{i=1}^n$	$d_i \times n$	Input data (n samples, d_0 features)
Weight (layer l)	$\mathbf{W}^{(l)}$	$d_{l+1} \times d_l$	Trainable parameters, $l = 0, \ldots, L$
Bias (layer l)	$\mathbf{b}^{(l)}$	$d_{l+1} \times 1$	Trainable parameters, broadcast to n samples
Pre-activation	$\mathbf{U}^{(l)}$	$d_{l+1} \times n$	$\mathbf{U}^{(l)} = \mathbf{W}^{(l)}\mathbf{H}^{(l)} + \mathbf{b}^{(l)}$
Activation	$\mathbf{H}^{(l)}$	$d_{l+1} \times n$	$\mathbf{H}^{(l)} = \Phi(\mathbf{U}^{(l)})$ (elementwise)
Loss	$\mathscr{L}$	scalar	Depends on $\mathbf{H}^{(L+1)} = \{\boldsymbol{X}_{\text{out}}^i\}_{i=1}^n$ and targets
Output error	$\boldsymbol{\delta}^{(L)}$	$d_o \times n$	$\boldsymbol{\delta}^{(L)} = \frac{\partial \mathscr{L}}{\partial \mathbf{H}^{(L+1)}} \odot \Phi'(\mathbf{U}^{(L)})$
Hidden error	$\boldsymbol{\delta}^{(l)}$	$d_{l+1} \times n$	$\boldsymbol{\delta}^{(l)} = \Phi'(\mathbf{U}^{(l)}) \odot \left((\mathbf{W}^{(l+1)})^T \cdot \boldsymbol{\delta}^{(l+1)}\right)$
Weight gradient	$\frac{\partial \mathscr{L}}{\partial \mathbf{W}^{(l)}}$	$d_{l+1} \times d_l$	$\frac{\partial \mathscr{L}}{\partial \mathbf{W}^{(l)}} = \boldsymbol{\delta}^{(l)} \cdot (\boldsymbol{H}^{(l)})^T$
Bias gradient	$\frac{\partial \mathscr{L}}{\partial \mathbf{b}^{(l)}}$	$d_{l+1} \times 1$	$\frac{\partial \mathscr{L}}{\partial \mathbf{b}^{(l)}} = \frac{1}{n}\sum_{i=1}^n \boldsymbol{\delta}_{i,:}^{(l)}$

tion:

- L: total number of layers (hidden).
- d_i: input dimension, d_o: output dimension, d_l: dimension of layer l.
- $\Phi(\cdot)$: activation function applied elementwise; Φ' its derivative.
- $\odot$: Hadamard (elementwise) product.
- $\cdot$: Matrix product.
- $\mathbf{W}^{(l)} \cdot \boldsymbol{\delta}^{(l)}$ is matrix multiplication ($d_l \times n$).
- $\boldsymbol{\delta}^{(l)} \cdot (\boldsymbol{H}^{(l)})^T$ is matrix multiplication ($d_{l+1} \times d_l$).
- For bias gradient, the sum over samples reduces to a column vector; in practice, one may use the raw sum (if the optimizer expects per-sample gradients) or average (as shown).

Note that in the last line of the table, we explicitly give the formula for the bias gradient, with consideration that

Chapter 7

Restricted Boltzmann Machines

Probabilistic generative models with a bipartite structure.

Chapter 8

Convolutional Neural Networks

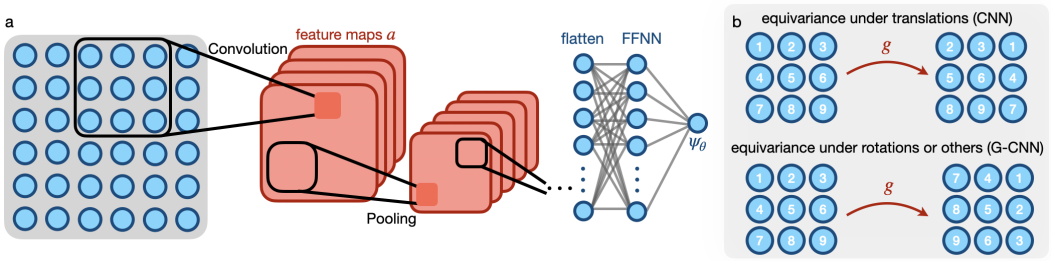


Figure 8.1: Structure of a Convolutional Neural Network

In MLP, the former layer is fully connected to the next layer, with each bond assigned with a weight matrix element. The number of training parameters thus scales with both sizes of two layers. Nevertheless, one may notice that in one data sample, the correlation among data points may not be that long, and it is possible to exploit this local correlation by using only short-ranged weight matrix mapping. At the same time, it can also be assumed that the data in one sample share some statistical similarity/homogeneity across itself, which makes the utilization of the same weight parameters across different locations in the input data feasible. Such a circumstance often happens in data set of images, or physical systems with high symmetries.

A convolutional neural network (CNN) employs a structure with convolutional layers, pooling layers, and fully connected layers, possibly with several circulations of the first two steps. A characteristic CNN structure is presented in Fig. **??**, where the input data is commonly some two-dimensional data like images, but equipped with multiple channels like RGB colors thus sharing data structure like $H \times W \times C_{\text{in}}$, with H and W denoting the height and width of the input data, and C_{in} denoting the number of input channels. The data is firstly organized to pass through convolutional layers where filters or kernels are applied to the input data of each convolutional layer. After each convolutional layer, a pooling layer is closely followed to renormalize or coarse-grain the data sample in its positional space. Such convolution-pooling operations are repeated multiple times to extract hierarchical features from the input data before feeding in an MLP. The process of channel passing can be parallelized with multiple channels.

In the usual structure, a convolutional layer contains C_{out} kernels K^β, $\beta = 1, 2, \ldots, C_{\text{out}}$, each of size $k \times k \times C_{\text{in}}$. We can split the kernel into representation $K^\beta = \{k_\alpha^\beta\}, \alpha = 1, 2, \ldots, C_{\text{in}}$, with each k_α^β a $k \times k$ matrix representing the weight matrix for the α-th input channel. Denoting the input data, or feature values as $X \in \mathbb{R}^{H \times W \times C_{\text{in}}}$, we also

split it into $X = \{x_\alpha\}, \alpha = 1, 2, \ldots, C_{\text{in}}$, with each $x_\alpha \in \mathbb{R}^{H\times W}$ representing the feature map for the α-th input channel. Notice that X can either be the original input data or the output from a previous layer. The output data $Y \in \mathbb{R}^{H'\times W'\times C_{\text{out}}}$ also reads $Y = \{y_\beta\}, \beta = 1, 2, \ldots, C_{\text{out}}$, with each $y_\beta \in \mathbb{R}^{H'\times W'}$ representing the feature map for the β-th output channel. For expressiveness, let us use positional coordinate to denote the matrix element $k_\alpha^\beta(i, j)$, $x_\alpha(i, j)$ and $y_\beta(i, j)$. The feature map then stands as

$$y_\beta(i,j) = \sum_{\alpha=1}^{C_{\text{in}}} [x_\alpha * k_\alpha^\beta](i,j) = \sum_{\alpha=1}^{C_{\text{in}}} \sum_{\tilde{i},\tilde{j}=0}^{k-1} x_\alpha(i+\tilde{i}, j+\tilde{j}) k_\alpha^\beta(\tilde{i},\tilde{j}), \tag{8.1}$$

with $*$ representing the convolution operation. There can also be a bias for each output channel b^β, and under this, we can write

$$y_\beta(i,j) = \sum_{\alpha=1}^{C_{\text{in}}} [x_\alpha * k_\alpha^\beta](i,j) + b^\beta, \tag{8.2}$$

or more compactly,

$$Y = K * X + \boldsymbol{b}, \tag{8.3}$$

with $K = \{k_\alpha^\beta\}$ a four-dimensional tensor and $\boldsymbol{b} = \{b^\beta\}$ broadcast to each output channel. Note that in representing the multiplication, we assign size

$$K : C_{\text{out}} \times C_{\text{in}} \times k \times k,\ Y : C_{\text{out}} \times H' \times W',\ X : C_{\text{in}} \times H \times W.$$

The output matrix size can change (shrink) when $k > 1$, as the convolution operation can reach the boundary of the input, where the kernel center can not reach the boundary, but one can always add zeros outside the true boundary (padding), or use some periodic boundary conditions so to make the size unchanged.

The output Y experience a further pooling operation to reduce its spatial dimensions, which is like a real space renormalization in physics. Basically, the output channel number of a pooling layer equals that of the input to that layer, so we just focus on each channel y_β. Not considering the pooling operation with coarse-grain window size $k_p \times k_p$:

$$\tilde{y}_\beta(i', j') = \text{pool}\left(\{y_\beta(i,j) | i = i's + u, j = j't + v,\ u, v = 0, 1, \ldots, k_p - 1\}\right), \tag{8.4}$$

and we may using *Max Pooling*

$$\text{pool}(\mathcal{S}) = \max_{s\in\mathcal{S}} s, \tag{8.5}$$

or *Average Pooling*

$$\text{pool}(\mathcal{S}) = \frac{1}{|\mathcal{S}|} \sum_{s\in\mathcal{S}} s, \tag{8.6}$$

to renormalize the data sample.

In physics, we may consider more sophisticated data structure aside the translational invariance. For instance, we may want to embed some group symmetry G like rotations in the aiming data, say wavefunction, structure, and a CNN embedded with group-equivariant convolution is called a group convolutional neural network (G-CNN), where the group convolution operation is defined as

$$[x * k](g) = \sum_{h\in G} x(gh) k(h), \tag{8.7}$$

in the sense that when the input and kernel are transformed under the group action, the convolution operation remains equivariant, i.e.,

$$[L_{g'}x * k](g) = \sum_{h\in G} x(gh)k(h) = \sum_{h\in G} x((g')^{-1}gh)k(h) = \sum_{l\in G} x(l)k(g^{-1}g'l),$$

$$L_{g'}[x * k][g] = [x * k][(g')^{-1}g] = \sum_{h\in G} x((g')^{-1}gh)k(h) = [L_{g'}x * k](g),$$

where the left action on the left side is defined.

Chapter 9

Recurrent Neural Networks

Memory and temporal dependencies.

In MLP, the information passes through the NN without cycles or memories, which means that a single input determines the whole output in a deterministic way in time. However, in our experience, our writing style is somehow like this, that we write a word, then another, then both of them determines how we write the next. This is somehow similar to the recurrent neural network (RNN): RNN has cycles and transmits information back into itself. This forms a series of interconnected states that capture temporal dependencies. Let us denote the input state at time step t as $\mathbf{X}_t \in \mathbb{R}^{d\times n}$, with n being the number of samples and d being the dimensionality of the input, and the hidden state at time step t as $\mathbf{H}_t \in \mathbb{R}^{h\times n}$, with h the number of hidden units. Then two weight matrices as input-to-hidden and hidden-to-hidden matrices can be defined as $\mathbf{W}_{xh} \in \mathbb{R}^{h\times d}$ and $\mathbf{W}_{hh} \in \mathbb{R}^{h\times h}$, respectively. By also defining hidden-to-output weight matrix $\mathbf{W}_{ho} \in \mathbb{R}^{o\times h}$, where o is the dimensionality of the output, we obtain the transmits equations

$$\mathbf{H}_t = \Phi(\mathbf{W}_{xh}\mathbf{X}_t + \mathbf{W}_{hh}\mathbf{H}_{t-1} + \mathbf{b}_h), \tag{9.1a}$$

$$\mathbf{O}_t = \Phi(\mathbf{W}_{ho}\mathbf{H}_t + \mathbf{b}_o), \tag{9.1b}$$

with $\mathbf{b}$ the broadcast bias vector. This three-layer RNN shares a simple structure

$$\begin{array}{ccccc}
\vdots & & \vdots & & \vdots \\
 & & \uparrow{\scriptstyle \mathbf{W}_{hh\cdot}} & & \\
\mathbf{X}_{t+1} & \xrightarrow{\mathbf{W}_{xh\cdot}} & \mathbf{H}_{t+1} & \xrightarrow{\mathbf{W}_{ho\cdot}} & \mathbf{O}_{t+1} \\
 & & \uparrow{\scriptstyle \mathbf{W}_{hh\cdot}} & & \\
\mathbf{X}_t & \xrightarrow{\mathbf{W}_{xh\cdot}} & \mathbf{H}_t & \xrightarrow{\mathbf{W}_{ho\cdot}} & \mathbf{O}_t \\
 & & \uparrow{\scriptstyle \mathbf{W}_{hh\cdot}} & & \\
\vdots & & \vdots & & \vdots
\end{array}, \tag{9.2}$$

with bias and activation function neglected.

This can be easily generalized to multiple hidden layer case, where we have $\mathbf{H}_t^{(l)} \in \mathbb{R}^{h_l\times n}$ for layer l, where h_l is the number of hidden units in layer $l = 1, 2, \cdots, L$. Correspondingly, we have intra-chain weight matrices like the MLP $\mathbf{W}_{l,l+1}^{(l)} \in \mathbb{R}^{h_{l+1}\times h_l}$ for layer l, where basically $l = 0, 1, \cdots, L$ with $l = 0$ for the input layer $\mathbf{X}_t$ and $h_0 = d$. We also have

inter-chain weight matrices counting on the recurrent process among neighboring time intervals $\mathbf{W}_{hh}^{(l)} \in \mathbb{R}^{h_l \times h_l}$ for layer l. They together form the transmits equations

$$\mathbf{H}_t^{(1)} = \Phi(\mathbf{W}_{xh}\mathbf{X}_t + \mathbf{W}_{hh}^{(1)}\mathbf{H}_{t-1}^{(1)} + \mathbf{b}_h^{(1)}), \tag{9.3a}$$

$$\mathbf{H}_t^{(l)} = \Phi(\mathbf{W}_{l-1,l}^{(l-1)}\mathbf{H}_t^{(l-1)} + \mathbf{W}_{hh}^{(l)}\mathbf{H}_{t-1}^{(l)} + \mathbf{b}_h^{(l)}),\ l = 2, \ldots, L-1, \tag{9.3b}$$

$$\mathbf{O}_t = \Phi(\mathbf{W}_{ho}\mathbf{H}_t^{(L)} + \mathbf{b}_o^{(L)}), \tag{9.3c}$$

where $\mathbf{W}_{xh} = \mathbf{W}_{01}^{(0)}$ and $\mathbf{W}_{ho} = \mathbf{W}_{L,L+1}^{(L)}$ are recognized. The 2D network structure of RNN can similarly be constructed as

$$\begin{array}{ccccccccccc}
\vdots & & \vdots & & \vdots & & \vdots & & \vdots \\
 & \mathbf{w}_{hh}^{(1)}\cdot\uparrow & & & \mathbf{w}_{hh}^{(l)}\cdot\uparrow & & & & \mathbf{w}_{hh}^{(L)}\cdot\uparrow & & \\
\mathbf{X}_{t+1} & \xrightarrow{\mathbf{w}_{xh}\cdot} & \mathbf{H}_{t+1}^{(1)} & \xrightarrow{\mathbf{w}_{12}^{(1)}\cdot} & \cdots & \xrightarrow{\mathbf{w}_{l-1,l}^{(l-1)}\cdot} & \mathbf{H}_{t+1}^{(l)} & \xrightarrow{\mathbf{w}_{l,l+1}^{(l)}\cdot} & \cdots & \xrightarrow{\mathbf{w}_{L-1,L}^{(L-1)}\cdot} & \mathbf{H}_{t+1}^{(L)} & \xrightarrow{\mathbf{w}_{ho}\cdot} & \mathbf{O}_{t+1} \\
 & \mathbf{w}_{hh}^{(1)}\cdot\uparrow & & & \mathbf{w}_{hh}^{(l)}\cdot\uparrow & & & & \mathbf{w}_{hh}^{(L)}\cdot\uparrow & & \\
\mathbf{X}_t & \xrightarrow{\mathbf{w}_{xh}\cdot} & \mathbf{H}_t^{(1)} & \xrightarrow{\mathbf{w}_{12}^{(1)}\cdot} & \cdots & \xrightarrow{\mathbf{w}_{l-1,l}^{(l-1)}\cdot} & \mathbf{H}_t^{(l)} & \xrightarrow{\mathbf{w}_{l,l+1}^{(l)}\cdot} & \cdots & \xrightarrow{\mathbf{w}_{L-1,L}^{(L-1)}\cdot} & \mathbf{H}_t^{(L)} & \xrightarrow{\mathbf{w}_{ho}\cdot} & \mathbf{O}_t \\
 & \mathbf{w}_{hh}^{(1)}\cdot\uparrow & & & \mathbf{w}_{hh}^{(l)}\cdot\uparrow & & & & \mathbf{w}_{hh}^{(L)}\cdot\uparrow & & \\
\mathbf{X}_{t-1} & \xrightarrow{\mathbf{w}_{xh}\cdot} & \mathbf{H}_{t-1}^{(1)} & \xrightarrow{\mathbf{w}_{12}^{(1)}\cdot} & \cdots & \xrightarrow{\mathbf{w}_{l-1,l}^{(l-1)}\cdot} & \mathbf{H}_{t-1}^{(l)} & \xrightarrow{\mathbf{w}_{l,l+1}^{(l)}\cdot} & \cdots & \xrightarrow{\mathbf{w}_{L-1,L}^{(L-1)}\cdot} & \mathbf{H}_{t-1}^{(L)} & \xrightarrow{\mathbf{w}_{ho}\cdot} & \mathbf{O}_{t-1} \\
 & \mathbf{w}_{hh}^{(1)}\cdot\uparrow & & & \mathbf{w}_{hh}^{(l)}\cdot\uparrow & & & & \mathbf{w}_{hh}^{(L)}\cdot\uparrow & & \\
\vdots & & \vdots & & \vdots & & \vdots & & \vdots
\end{array}, \tag{9.4}$$

The output of an RNN can be the last hidden state $\mathbf{H}_T^{(L)}$, the last output $\mathbf{O}_T$, all outputs $\mathbf{O}_t$ in either averaged or separate way, etc., depending on the specific task and circumstance. Thus the supervised learning of RNN involves optimizing loss function in the general form

$$\mathscr{L} = \sum_{t=1}^{T} \ell(\mathbf{O}_t, \mathbf{Y}_t), \tag{9.5}$$

where $\ell(\cdot, \cdot)$ is a suitable loss function and $\mathbf{Y}_t$ is the true label at time t.

Chapter 10

Transformer

Attention mechanism and self-attention.

Chapter 11

Neural Quantum States

11.1 Representation of quantum states

We may expand the target wavefunction $|\Psi\rangle$ over a complete orthonormal basis $|\sigma\rangle$ (say a configurational space):

$$|\Psi\rangle = \sum_{\sigma} \psi(\sigma) \, |\sigma\rangle \, , \tag{11.1}$$

with $\psi(\sigma) = \langle\sigma|\Psi\rangle$ the amplitude of the wavefunction in the σ basis. Notice that generally $\psi(\sigma)$ is our target and is unknown. Now we would like to learn $\psi(\sigma)$ from some deep learning structure, then by denoting θ the parameters of the neural network, we can write

$$|\Psi_\theta\rangle = \sum_{\sigma} \psi_\theta(\sigma) \, |\sigma\rangle \, , \tag{11.2}$$

and this state is known as the neural quantum state (NQS).

11.2 Beyond Euclidean gradient descent

The Euclidean gradient descent relies on energy surface descent over gradient direction curve. With well-defined energy surface and chosen well-behaved activation function, the usual gradient descent can always lead us towards some (local) minimum. Nevertheless, this set of calculation is based on the optimization of the loss function — the free energy here, but since we focus more on the wavefunction (components) in quantum mechanics, it will be (at least in principle) more natural to directly guide how the wavefunction is optimized under the gradient.

For energy surface gradient, we utilize the Euclidean metric to measure the descent on/near the energy surface, and to generalize it to some wavefunction gradient, we need to introduce the metric over space where quantum state live — the normalized Hilbert space, where the distance between the two states is described by the Fubini-Study distance/quantum fidelity

$$F(\Psi, \Phi) = \cos\theta(\Psi, \Phi) = |\, \langle\Psi|\Phi\rangle \,|_{\text{Norm}} = \sqrt{\frac{\langle\Psi|\Phi\rangle \, \langle\Phi|\Psi\rangle}{\langle\Psi|\Psi\rangle \, \langle\Phi|\Phi\rangle}}. \tag{11.3}$$

For NQS, we learn by slowly varying the training parameters θ, which makes it possible to

define a smooth variation of quantum states, and introduce the so-called quantum metric

$$F(\Psi_\theta, \Psi_{\theta+\delta\theta}) = \sum_{i,j} g_{ij}(\theta)\delta\theta_i\delta\theta_j, \tag{11.4}$$

where for simplicity we utilize one index to label all NN parameters.

11.3 Topological configuration in training

Chapter 12

Physics Informed Machine Learning